

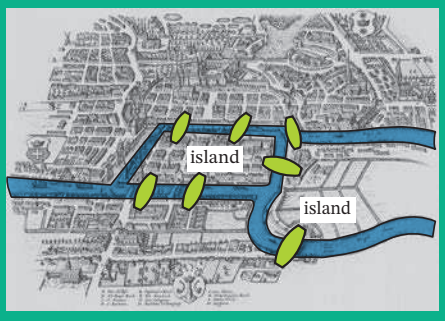
6

MEASUREMENT AND SPACE

GEOMETRY AND NETWORKS

During the 18th century, in the Prussian town of Königsberg (now Kaliningrad, Russia, pictured), there were 7 bridges connecting 2 islands and the banks of the Pregel River (see map below). There was a tradition on Sundays where people would walk through the town try to complete a walk crossing each bridge only once. No person was able to do it but no one could prove that it was impossible.

In 1736, the Swiss mathematician Leonhard Euler solved the problem by reducing the map to a network diagram, creating a new field of mathematics called **graph theory**. Euler proved that it was impossible to do the walk. Eventually, another bridge was built.



Alamy Stock Photo/Album



Chapter outline

	Working mathematically				
6.01 Triangle geometry	U	F		R	C
6.02 Quadrilateral geometry	U	F		R	C
6.03 Angle sum of a polygon*	U	F		R	C
6.04 Exterior angle sum of a convex polygon*	U	F		R	C
6.05 Line and rotational symmetry*	U	F		R	C
6.06 Transformations on the number plane*	U	F		R	C
6.07 Networks*	U	F		R	C
6.08 Planar graphs and Euler's formula*	U	F		R	C

***PATH**

Wordbank

angle sum The total of the sizes of all the angles in a shape, such as a triangle

convex quadrilateral A quadrilateral whose vertices all point outwards

diagonal An interval joining 2 non-adjacent vertices of a shape

edge (of a network) A line that connects vertices (points) together

exterior angle of a triangle An 'outside' angle of a triangle formed after extending one of the sides of the triangle

planar graph A connected network (graph) where the edges do not cross (overlap)

regular polygon A polygon with all angles equal and all sides equal, such as a square

transformation Translation (slide), reflection (flip) or rotation (turn)

vertex (of a network) A point representing an item that is connected by lines called edges.

Quiz
Wordbank 6

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (5):

6.01 Angle sum of a triangle • Exterior angle of a triangle 1 • Exterior angle of a triangle 2

6.07 Networks

6.08 Euler's formula for planar graphs

Twig videos (2):

6.07 The seven bridges of Konigsberg • Networks: Labyrinths and mazes

Quizzes (5):

- Wordbank 6
- SkillCheck 6
- Mental skills 6
- Language of maths 6
- Test yourself 6

Skillsheets (2):

6.01 Naming shapes

6.05 Line and rotational symmetry

Worksheets (15):

6.02 Classifying quadrilaterals • Properties of triangles and quadrilaterals • Naming quadrilaterals • Deductive geometry • Diagonal properties of quadrilaterals • Shapes and angles review

6.03 Angle sum of a polygon • Find the unknown angle • Equal angles

6.04 Angles in polygons

6.05 Symmetry • Symmetry of flat shapes

6.06 Transformations • 5 mm grid paper

Mind map Geometry and networks (Advanced)

Puzzles (3):

6.02 Mixed angle problems • Parallel lines • Triangles and quadrilaterals

Presentation (1):

6.02 Angles and shapes

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

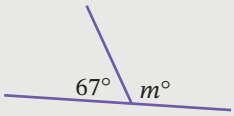
In this chapter you will:

- classify triangles, quadrilaterals and polygons
- solve geometrical problems involving the angle sum of a triangle and quadrilateral and the exterior angle of a triangle
- (PATH) solve geometrical problems involving the angle sum of a polygon and the exterior angle sum of a convex polygon
- (PATH) identify line and rotational symmetries in plane shapes
- (PATH) translate, reflect and rotate points and shapes on a number plane using coordinates, where the reflections are across the x- or y- axis and the rotations through multiples of 90° about the origin
- (PATH) describe networks using appropriate terminology, including vertex, edge, path and degree
- (PATH) describe a planar graph and apply Euler's formula

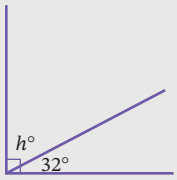
SkillCheck ANSWERS ON P. 625

1 Find the value of each variable.

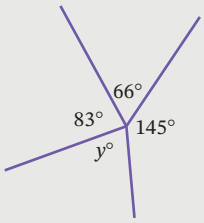
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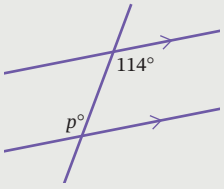
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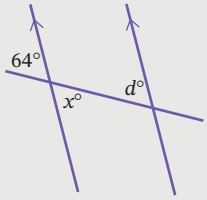
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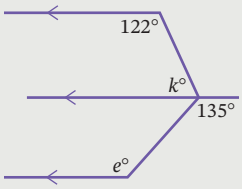
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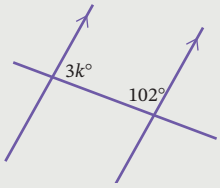
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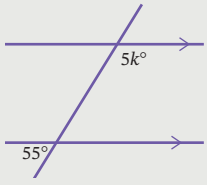
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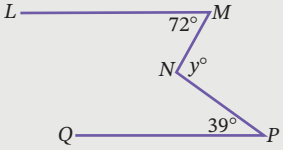
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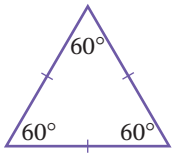
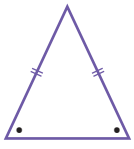
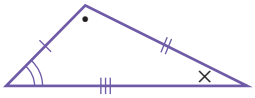
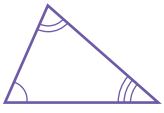
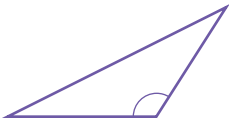
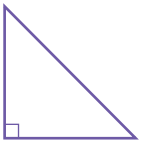


i



Triangle geometry 6.01

Triangles can be classified in 2 ways: by their sides or by their angles.

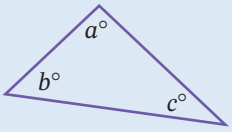
Classifying by sides		
Equilateral triangle	**Isosceles triangle**	**Scalene triangle**
 3 equal sides (Also 3 equal angles, each 60°)	 2 equal sides (Also 2 equal angles, opposite the equal sides)	 No equal sides (Also no equal angles)
Classifying by angles		
Acute-angled triangle	**Obtuse-angled triangle**	**Right-angled triangle**
 3 acute angles (less than 90°)	 One obtuse angle (between 90° and 180°)	 One right angle (90°)

Video
Angle sum of a triangle

Angle sum of a triangle

The **angle sum** of a triangle is 180°.

$a + b + c = 180$



Proof:

Consider any triangle PQR with angles $a°$, $b°$ and $c°$.

Construct a line through P parallel to QR.

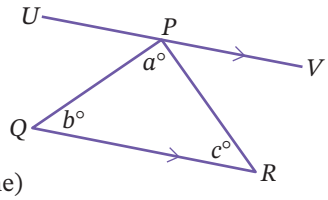
$\therefore \angle UPQ = b°$ (alternate angles, $UV \parallel QR$)

and $\angle VPR = c°$ (alternate angles, $UV \parallel QR$)

But $\angle UPQ + \angle QPR + \angle VPR = 180°$ (angles on a straight line)

$\therefore a° + b° + c° = 180°$

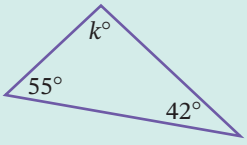
$\therefore$ The sum of the angles of a triangle is 180°.



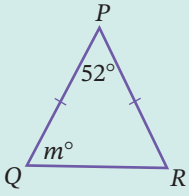
Example 1

Find the value of each variable, giving reasons.

a



b



Solution

a $k + 55 + 42 = 180$ (angle sum of a triangle)

$k = 180 - 55 - 42$

$= 83$

b $\angle R = \angle Q = m°$ ($\triangle PQR$ is isosceles)

$m + m + 52 = 180$ (angle sum of a triangle)

$2m + 52 = 180$

$2m = 128$

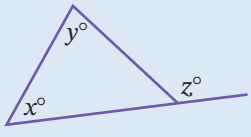
$m = 64$

Video
Exterior angle of a triangle 1

The exterior angle of a triangle

The **exterior angle** of a triangle is equal to the sum of the 2 interior opposite angles.

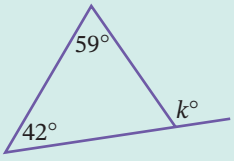
$z = x + y$



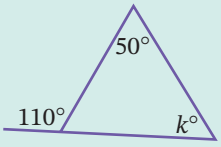
Example 2

Find the value of k in each diagram.

a



b



Solution

a $k = 42 + 59$ (exterior angle of triangle)
$= 101$

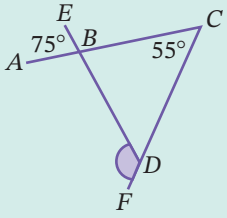
b $k + 50 = 110$ (exterior angle of triangle)
$k = 60$

Video
Exterior angle of a triangle 2

6.01

Example 3

Find the size of the marked angle $\angle BDF$.



Solution

$\angle DBC = \angle ABE = 75°$ (vertically opposite angles)
$\angle BDF = 75° + 55°$ (exterior angle of $\triangle BCD$)
$= 130°$

Video
Exterior angle of a triangle 2

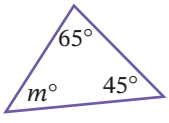
EXERCISE 6.01 ANSWERS ON P. 625

Triangle geometry U F R C

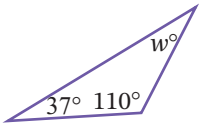
1 Find the value of each variable, giving reasons. R C

EXAMPLE 1

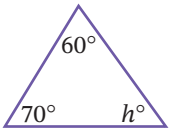
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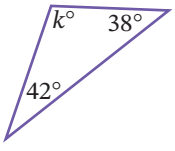
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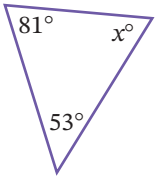
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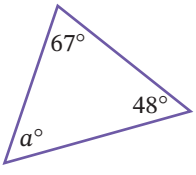
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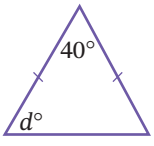
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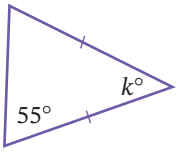
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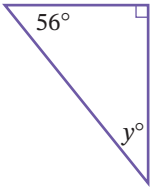
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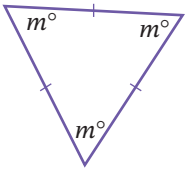
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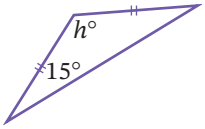
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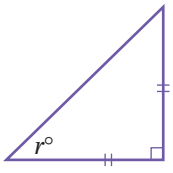
j



k



l



Foundation | Standard | Complex

2 What is the size of each angle in an equilateral triangle? Select **A**, **B**, **C** or **D**.

A 30° **B** 45° **C** 60° **D** 90°

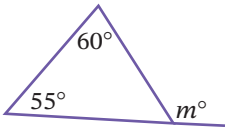
3 One angle of an isosceles triangle is equal to 80°. Which 2 of the following could be the sizes of the other angles? Select 2 of **A**, **B**, **C** or **D**. R

A 80° and 20° **B** 40° and 60° **C** 40° and 40° **D** 50° and 50°

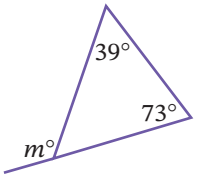
EXAMPLE 2

4 Find the value of m. R C

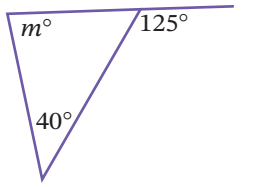
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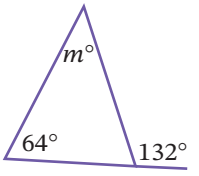
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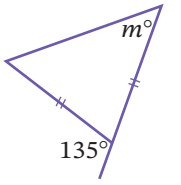
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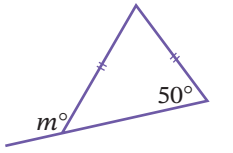
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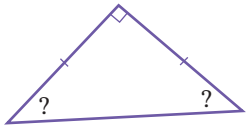
e



f



5 What is the size of the unknown angles in this triangle? Select **A**, **B**, **C** or **D**. R



A 30° **B** 45°

C 60° **D** 90°

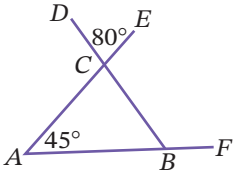
6 One angle of an obtuse-angled triangle is 50°. Which of the following could be the sizes of the other angles? Select **A**, **B**, **C** or **D**. R

A 80° and 50° **B** 100° and 30° **C** 65° and 65° **D** 45° and 95°

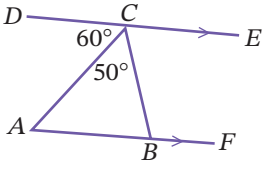
EXAMPLE 3

7 Find the size of $\angle CBF$ in each diagram. R C

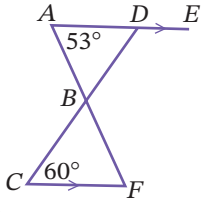
a



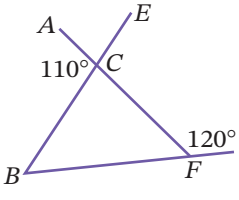
b



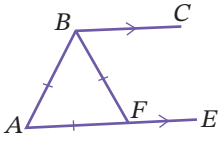
c



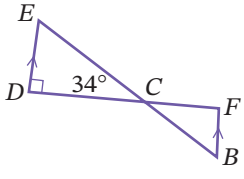
d



e



f

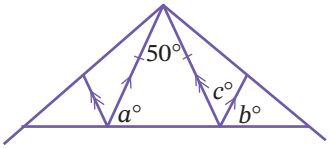


Foundation Standard Complex

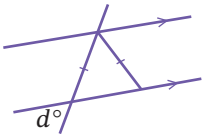
9780170479585

8 An isosceles triangle has a side length of 6 cm and one of its angles equal to 40°. Draw all possible shapes of the triangle. R

9 The diagram shows the shape of a roof truss. Find the value of each variable. R

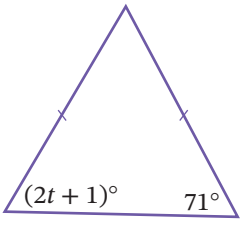


10 Copy this diagram and mark all angles equal to the angle marked $d°$. R

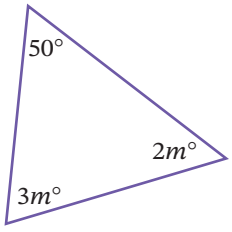


11 Find the value of each variable. R C

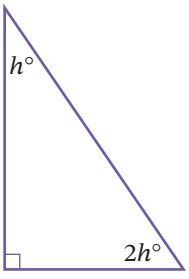
a



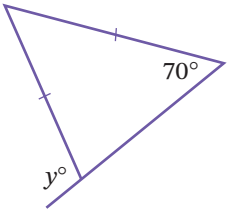
b



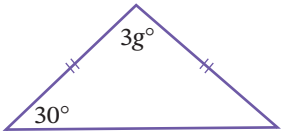
c



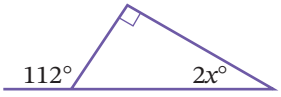
d



e

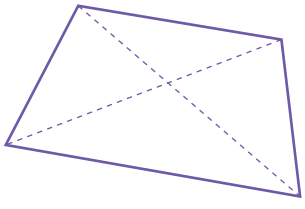
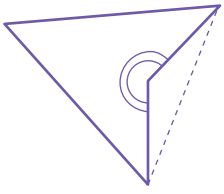


f



Quadrilateral geometry 6.02

A **quadrilateral** is any shape with 4 straight sides. A quadrilateral may be either **convex** or **non-convex** (**concave**).

Convex quadrilateral	Non-convex quadrilateral
	
• All **vertices** (corners) point outwards. • All **diagonals** lie within the shape. • All angles are less than 180°.	• One vertex points inwards. • One diagonal lies outside the shape. • One angle is more than 180° (reflex angle).

Worksheets
Classifying quadrilaterals
Properties of triangles and quadrilaterals
Naming quadrilaterals
Deductive geometry
Diagonal properties of quadrilaterals
Shapes and angles review

Foundation Standard Complex

Skillsheet
Naming shapes

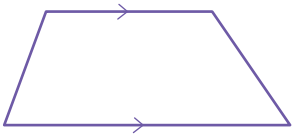
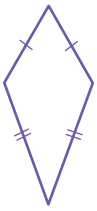
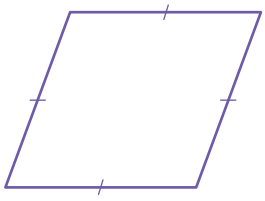
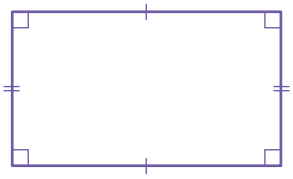
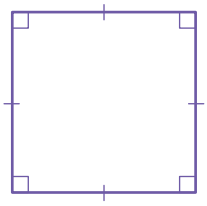
Puzzles
Mixed angle problems

Parallel lines

Triangles and quadrilaterals

Presentation
Angles and shapes

There are 6 special types of quadrilaterals.

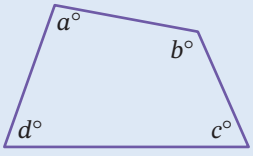
Trapezium	Kite	Parallelogram
		
One pair of parallel sides	2 pairs of equal adjacent sides 'Adjacent' means 'next to each other'.	2 pairs of parallel sides
Rhombus	**Rectangle**	**Square**
		
4 equal sides	4 right angles	4 equal sides and 4 right angles

Angle sum of a quadrilateral

The **angle sum of a quadrilateral** is 360°.

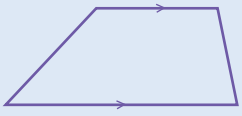
$a + b + c + d = 360$

This property is true for both convex and non-convex quadrilaterals.



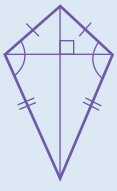
Properties of quadrilaterals

Trapezium



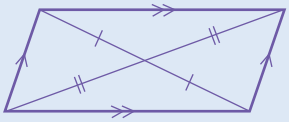
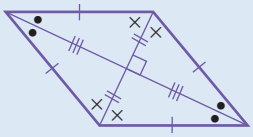
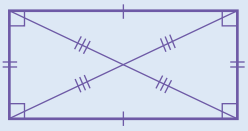
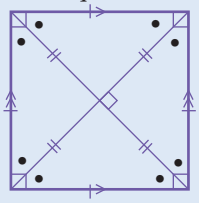
- One pair of opposite sides parallel
- No axes of symmetry

Kite



- 2 pairs of equal adjacent sides
- One pair of opposite angles equal
- One axis of symmetry
- Diagonals intersect at right angles

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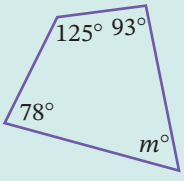
Parallelogram 	• Opposite sides are parallel and equal • Opposite angles are equal • No axes of symmetry • Diagonals bisect each other
Rhombus 	• All sides are equal • 2 axes of symmetry • A special type of parallelogram • Diagonals bisect each other at right angles • Diagonals bisect the angles of the rhombus
Rectangle 	• All angles are 90° (right angles) • 2 axes of symmetry • A special type of parallelogram • Diagonals are equal (in length) • Diagonals bisect each other
Square 	• All sides are equal • All angles are 90° (right angles) • 4 axes of symmetry • A special type of rhombus and rectangle • Diagonals are equal • Diagonals bisect each other at right angles • Diagonals bisect the angles of the square

6.02

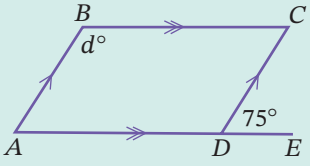
Example 4

Find the value of each variable.

a



b



Solution

a $m + 78 + 125 + 93 = 360$ (angle sum of a quadrilateral)

$m = 360 - 78 - 125 - 93$

$= 64$

b $\angle CDA = 180° - 75°$ (angles on a straight line)

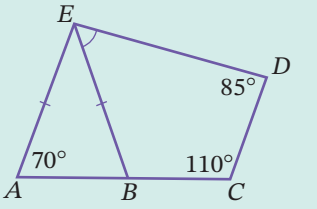
$= 105°$

$d = 105$ (opposite angles of a parallelogram)

Can you see another method for finding d?

Example 5

Find the size of $\angle BED$.



Solution

$\angle ABE = 70°$ (equal angles in isosceles $\triangle ABE$)

$\angle EBC = 180° - 70°$ (angles on a straight line)

$= 110°$

$\angle BED = 360° - 85° - 110° - 110°$ (angle sum of quadrilateral $BCDE$)

$= 55°$

Can you see another method for finding $\angle BED$?

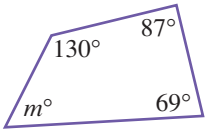
EXERCISE 6.02 ANSWERS ON P. 625

Quadrilateral geometry U F R C

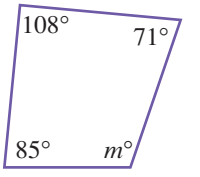
EXAMPLE 4

1 Find the value of m in each diagram. R

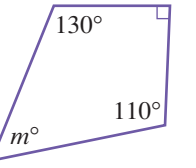
a



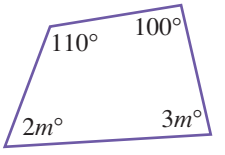
b



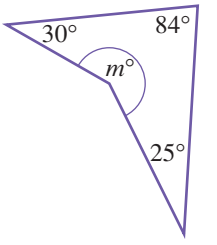
c



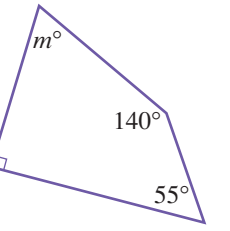
d



e



f



2 Copy and complete this table. R C

Property	Trapezium	Kite	Parallelogram	Rhombus	Rectangle	Square
One pair of opposite sides parallel						
Opposite sides parallel						
Opposite sides equal			✓			
All sides equal						
2 pairs of adjacent sides equal						
Diagonals equal					✓	

Foundation Standard Complex

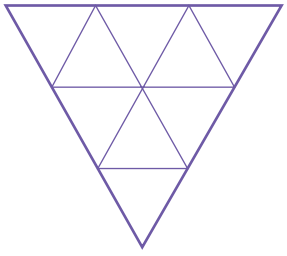
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Property	Trapezium	Kite	Parallelogram	Rhombus	Rectangle	Square
Diagonals bisect each other						
Diagonals meet at right angles						
Diagonals bisect the angles of the shape						
Opposite angles equal						
One pair of opposite angles equal						
All angles 90°						
Axes of symmetry	0					

3 Refer to the table you completed in question **2**, and name all quadrilaterals that have: R C

a no axes of symmetry
b one pair of parallel sides
c 4 equal sides
d equal diagonals
e opposite sides equal
f 4 axes of symmetry
g adjacent sides equal
h one axis of symmetry
i opposite sides parallel
j all angles measuring 90°
k 2 axes of symmetry
l diagonals which bisect each other
m opposite angles equal
n diagonals meeting at 90°

4 Each triangle in this diagram is an equilateral triangle. R



a Name the different types of quadrilaterals that you can find in the diagram.
b How many of each type of quadrilateral are there in the diagram?
c How many triangles are there? (It may help to copy the diagram so that you can draw on it using coloured pencils.)

5 Which statements are *always* true? R

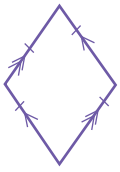
A A rhombus is a parallelogram.
B The diagonals of a parallelogram meet at right angles.
C A square is a rhombus.
D A parallelogram is a quadrilateral with a pair of opposite sides equal and parallel.
E A square is a rectangle.
F The diagonals of an isosceles trapezium bisect each other.
G The opposite angles of a rhombus are equal.
H The diagonals of a rhombus are equal and bisect each other at right angles.
I A rectangle is a parallelogram.

6 Name each quadrilateral using the properties marked on it. R

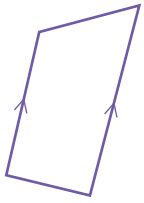
a



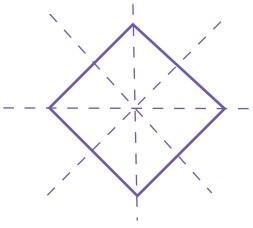
b



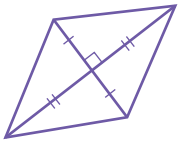
c



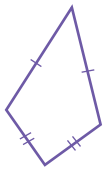
d



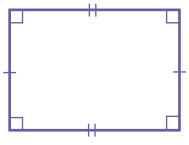
e



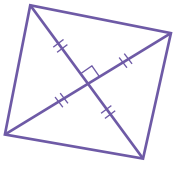
f



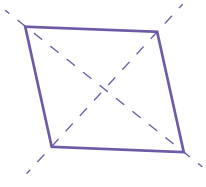
g



h

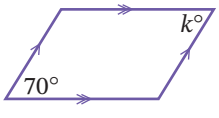


i

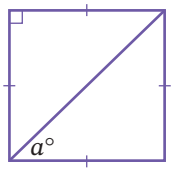


7 Find the value of each variable, giving reasons. R C

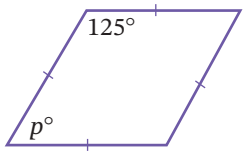
a



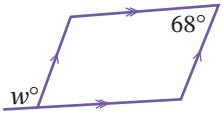
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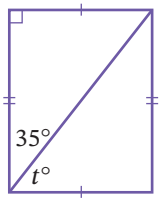
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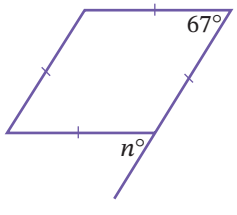
d



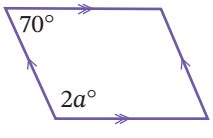
e



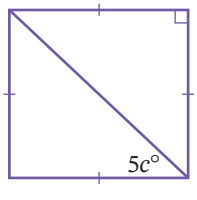
f



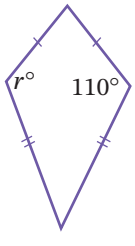
g



h



i



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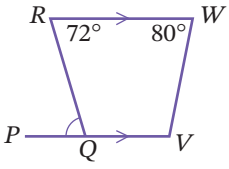
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EXAMPLE 5

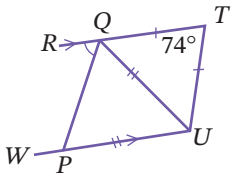
6.02

8 Find the size of $\angle PQR$ in each diagram, giving reasons. R C

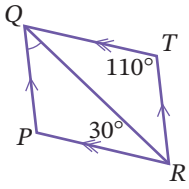
a



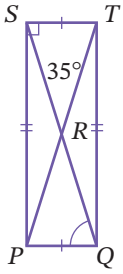
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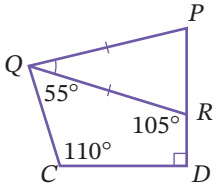
c



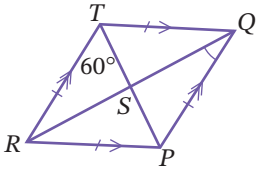
d



e



f



9 Use geometrical instruments or dynamic geometry software to construct each quadrilateral.

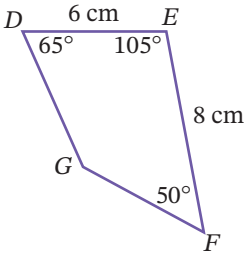
a Parallelogram $TUVW$ with sides $TU = 30$ mm, $UV = 60$ mm and $\angle TUV = 113°$.

b Rhombus with side length 5 cm.

c Convex quadrilateral $WXYZ$ with $WX = 65$ mm, $WZ = 40$ mm, $\angle ZWX = 54°$, $\angle WZY = 114°$ and $XY = 24$ mm.

d Non-convex quadrilateral with one side 4.5 cm and one angle 200°.

e



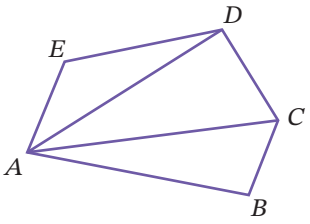
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Investigation

Angle sum of a polygon

To find the angle sum of a pentagon *ABCDE* (5 sides), follow this reasoning.

From one vertex of the pentagon, draw all the diagonals. The diagonals from vertex *A* have divided the pentagon into 3 triangles.



$\therefore$ Angle sum of a pentagon = angle sum of the 3 triangles

$= 3 \times 180°$

$= 540°$

1 **a** Draw a hexagon (6 sides) and from one vertex draw all the diagonals.

b How many diagonals are there?

c How many triangles did you form?

d Hence find the angle sum of a hexagon.

2 Repeat the procedure to find the angle sum of:

a an octagon (8 sides) **b** a decagon (10 sides)

3 Copy and complete this table.

Polygon	Number of sides	Number of triangles	Angle sum
triangle	3	1	180°
quadrilateral	4	2	
pentagon	5		
hexagon			
octagon			
decagon			

4 Copy and complete this pattern: The number of triangles formed is always 2 _____ than the number of _____ of the polygon.

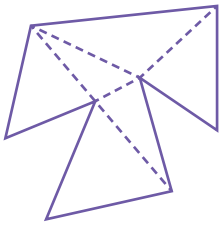
5 **a** Using your own words, describe the rule for finding the angle sum of a polygon.

b What is the angle sum of a polygon with 20 sides?

c For a polygon with n sides, write a formula for the sum of its angles. Discuss your result with other students.

6 **a** Draw a non-convex octagon.

b Divide the polygon into triangles as shown.



c How many triangles have been formed?

d Find the angle sum of this non-convex octagon.

e Does your rule for the angle sum of a polygon also apply to the non-convex polygon?

Angle sum of a polygon

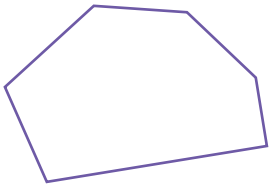
6.03

PATH

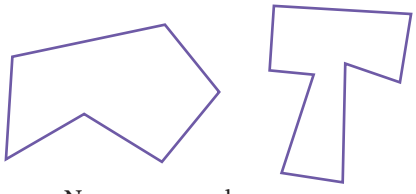
6.03

Worksheet
Angle sum of a polygon

A **polygon** is a general name for any shape with straight sides. The word is derived from the Greek, meaning 'many angles'. Shapes with curved sides, such as circles, ellipses and semicircles, are *not* polygons.



Convex polygon



Non-convex polygons

In a non-convex polygon, some vertices point inwards, some diagonals lie outside the shape and some angles are more than 180° (reflex angles).

Name	Number of sides
Pentagon	5
Hexagon	6
Heptagon	7
Octagon	8
Nonagon	9
Decagon	10
Undecagon	11
Dodecagon	12

Angle sum of a polygon

The **angle sum of a polygon** with n sides is given by the formula $A = 180(n - 2)°$.

This formula applies to both convex and non-convex polygons.

Example 6

Find the angle sum of a nonagon (9 sides).

Solution

Substitute $n = 9$ into $A = 180(n - 2)°$.

Angle sum $= 180(9 - 2)°$

$= (180 \times 7)°$

$= 1260°$

PATH

Example 7

Find the number of sides of a polygon that has an angle sum of 720°.

Solution

$180(n-2) = 720 \quad A = 720$

$180n - 360 = 720$

$180n = 1080$

$n = \frac{1080}{180}$

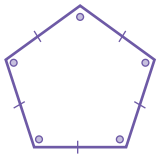
$= 6$

$\therefore$ The polygon has 6 sides (hexagon).

Worksheet
Equal angles

Regular polygons

A **regular polygon** has all angles equal and all sides equal. For example, a regular pentagon has 5 equal sides and 5 equal angles. A square is a regular polygon but a rhombus is not.



Angle in a regular polygon

The size of each angle in a **regular polygon** with n sides is

$$\frac{\text{Angle sum}}{\text{No. of sides}} = \frac{180(n-2)^\circ}{n}$$

Example 8

Find the size of one angle in a regular hexagon.

Solution

A hexagon has 6 sides ($n = 6$).

Size of one angle $= \frac{180(6-2)^\circ}{6}$

$= \frac{(180 \times 4)^\circ}{6}$

$= 120^\circ$

Each angle in a regular hexagon is 120°.

EXERCISE 6.03 ANSWERS ON P. 626

Angle sum of a polygon U F R C

1 How many sides has:

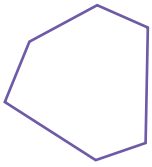
a a hexagon? **b** a quadrilateral? **c** a nonagon?

d a decagon? **e** a heptagon? **f** a pentagon?

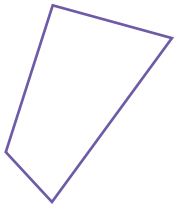
g a dodecagon? **h** an octagon? **i** an undecagon?

2 Name each polygon. R C

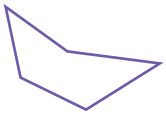
a



b



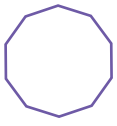
c



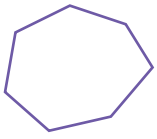
d



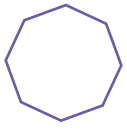
e



f



g



h

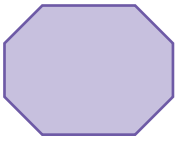


3 Which polygons from question **2** are: R C

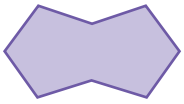
a convex? **b** regular? **c** concave?

4 Which shape is a regular octagon? Select **A**, **B**, **C** or **D**.

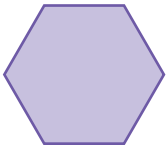
A



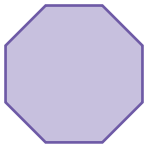
B



C



D



5 What is the more common name for: R C

a a regular triangle? **b** a regular quadrilateral?

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EXAMPLE 6

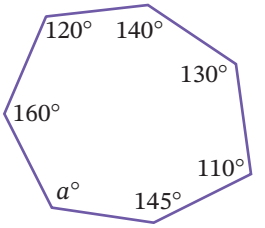
6 Find the angle sum of a polygon with:

a 5 sides **b** 8 sides **c** 15 sides **d** 12 sides

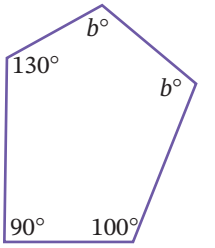
e 7 sides **f** 10 sides **g** 20 sides **h** 11 sides

7 Find the value of each variable. R

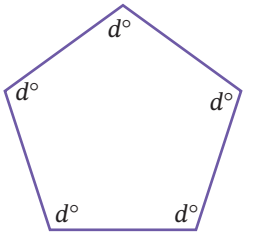
a



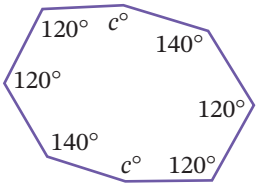
b



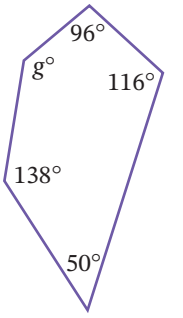
c



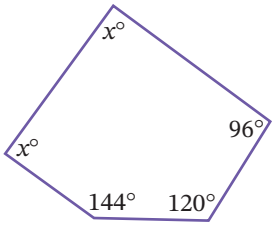
d



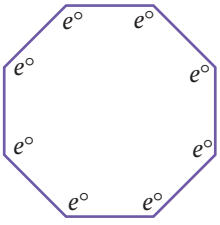
e



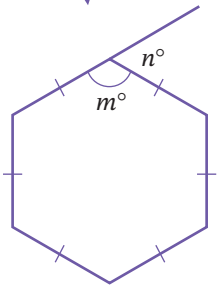
f



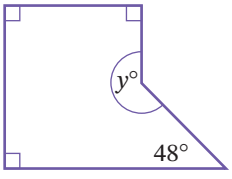
g



h



i



EXAMPLE 7

8 Find the number of sides of a polygon that has an angle sum of: R

a 2160° **b** 1620° **c** 3960° **d** 2700°

9 Which polygon has an angle sum of 1440°? Select **A**, **B**, **C** or **D**. R

A pentagon **B** decagon **C** nonagon **D** octagon

EXAMPLE 8

10 Find the size of one angle in a regular: R

a octagon **b** decagon **c** dodecagon **d** hexagon

11 The angle sum of a regular polygon is 6840°. R

a How many sides does the polygon have?

b Find the size of each angle.

12 How many sides does a regular polygon have if each of its angles is: R

a 165°? **b** 170°? **c** 144°?

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Mental skills 6: Maths without calculators ANSWERS ON P. 626

Quiz
Mental skills 6

Dividing decimals

To divide one decimal by another, first move the decimal points in both decimals the same number of places to the right so that the second decimal is a whole number.

1 Study each example.

a $0.24 \div 0.06 = 24 \div 6 = 4$

b $0.45 \div 0.5 = 4.5 \div 5 = 0.9$

c $0.006 \div 0.3 = 0.06 \div 3 = 0.02$

d $27 \div 0.9 = 270 \div 9 = 30$

e $1.6 \div 0.4 = 16 \div 4 = 4$

f $5.6 \div 0.07 = 560 \div 7 = 80$

2 Now evaluate each quotient.

a $0.25 \div 0.5$ **b** $63 \div 0.7$ **c** $3.2 \div 0.4$ **d** $0.18 \div 0.2$

e $2.7 \div 0.03$ **f** $0.042 \div 0.06$ **g** $4 \div 0.5$ **h** $1.2 \div 0.04$

i $0.072 \div 0.9$ **j** $0.35 \div 0.1$ **k** $0.28 \div 0.07$ **l** $0.033 \div 0.11$

3 Study each example.

Given that $112 \div 14 = 8$, evaluate each expression.

a
$$\begin{aligned} 112 \div 1.4 &= 112.0 \div 1.4 \\ &= 1120 \div 14 \\ &= 112 \times 10 \div 14 \\ &= 112 \div 14 \times 10 \\ &= 8 \times 10 \\ &= 80 \end{aligned}$$

Estimate: $112 \div 1.4 \approx 112 \div 1 = 112$

b
$$\begin{aligned} 0.112 \div 0.14 &= 0.112 \div 0.14 \\ &= 11.2 \div 14 \\ &= 112 \div 10 \div 14 \\ &= 112 \div 14 \div 10 \\ &= 8 \div 10 \\ &= 0.8 \end{aligned}$$

Estimate: $0.112 \div 0.14 \approx 0.1 \div 0.1 = 1$

c
$$\begin{aligned} 1120 \div 1.4 &= 11\,200 \div 14 \\ &= 112 \times 100 \div 14 \\ &= 112 \div 14 \times 100 \\ &= 8 \times 100 \\ &= 800 \end{aligned}$$

Estimate: $1120 \div 1.4 \approx 1120 \div 1 = 1120$

d
$$\begin{aligned} 1.12 \div 14 &= 112 \div 100 \div 14 \\ &= 112 \div 14 \div 100 \\ &= 8 \div 100 \\ &= 0.08 \end{aligned}$$

Estimate: $1.12 \div 14 \approx 1.12 \div 10 = 0.112$

4 Now given that $368 \div 23 = 16$, evaluate each quotient.

a $36.8 \div 2.3$ **b** $368 \div 2.3$ **c** $3.68 \div 2.3$ **d** $0.368 \div 0.23$

e $36.8 \div 23$ **f** $3.68 \div 0.23$ **g** $36.8 \div 0.23$ **h** $0.368 \div 2.3$

I $0.368 \div 23$ **j** $3.68 \div 0.023$ **k** $3.68 \div 23$ **l** $0.368 \div 230$

PATH

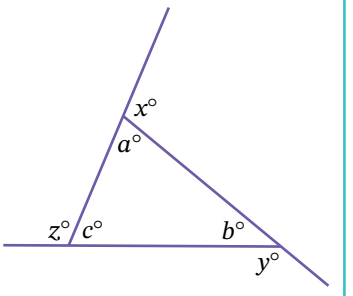
Investigation

Exterior angle sum of a convex polygon

1 **a** Draw a triangle and extend each side to show the exterior angles of the triangle as shown.

Exterior angles are $x°$, $y°$ and $z°$.

Interior angles are $a°$, $b°$ and $c°$.



b Use a protractor to measure angles $x°$, $y°$ and $z°$.

c Find the sum of the exterior angles.

d Looking at the diagram, what must be the value of $a° + x° + b° + y° + c° + z°$?

e But what do we know about the value of $a° + b° + c°$?

f Therefore, what must be the value of $x° + y° + z°$?

2 Repeat this procedure for the exterior angles of a convex quadrilateral. What is the sum of the exterior angles of a convex quadrilateral?

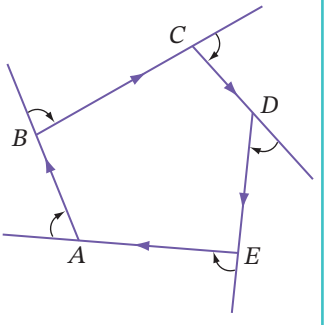
3 Repeat the procedure for a convex pentagon and a convex hexagon. What do you notice about the sum of the exterior angles of those polygons?

4 **a** Draw any convex polygon and extend the sides. Label the vertices of your polygon A, B, C, etc.

b Start at A and move around the polygon, turning in the direction indicated at each vertex.

c Continue until you return to A, facing the same way you started. What must be the sum of the turns in any round trip of a convex polygon?

d Test whether this rule works for a non-convex polygon.



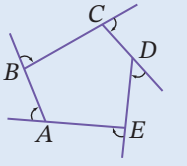
6.04 Exterior angle sum of a convex polygon

PATH

Worksheet
Angles in polygons

Exterior angle sum of a convex polygon

The **sum of the exterior angles** of a convex polygon is 360°.

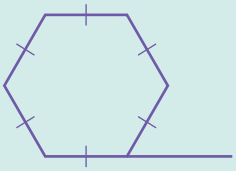


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Example 9

For a regular hexagon, find the size of:

a each exterior angle **b** each (interior) angle.



Solution

a Sum of exterior angles = 360°

One exterior angle = 360° ÷ 6 = 60°

b Each angle = 180° − 60° = 120° OR: Each angle $= \frac{180(6-2)^\circ}{6} = 120^\circ$

Example 10

Find the number of sides of a regular polygon if:

a each exterior angle is 12° **b** each (interior) angle is 160°.

Solution

a Sum of exterior angles = 360°

Number of exterior angles = 360 ÷ 12 = 30

∴ The regular polygon has 30 sides.

b Exterior angle = 180° − 160° = 20° (angles on a straight line)

Sum of exterior angles = 360°

Number of exterior angles = 360 ÷ 20 = 18

∴ The regular polygon has 18 sides.

OR: $\frac{180(n-2)^\circ}{n} = 160^\circ$

$180(n-2) = 160n$

$180n - 360 = 160n$

$20n - 360 = 0$

$20n = 360$

$n = \frac{360}{20} = 18$

∴ The regular polygon has 18 sides.

EXERCISE 6.04 ANSWERS ON P. 626

Exterior angle sum of a convex polygon U F R

EXAMPLE 9

1 Find the size of each exterior angle of a regular:

a octagon **b** decagon **c** 15-sided polygon **d** nonagon

2 Find the size of each angle of a regular: R

a pentagon **b** dodecagon **c** nonagon **d** 16-sided polygon

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PATH

EXAMPLE 10

3 Find the number of sides of a regular polygon if each exterior angle is:

a 24° **b** 36° **c** 40°

d 10° **e** 18° **f** 60°

4 Find the number of sides of a regular polygon if each angle is: R

a 150° **b** 175° **c** 162°

d 140° **e** 108° **f** 176°

6.05 Line and rotational symmetry

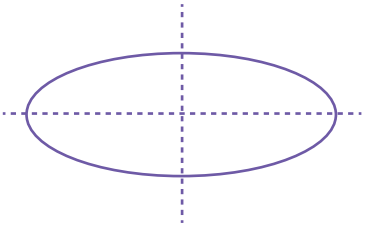
PATH

Skillsheet
Line and rotational symmetry'

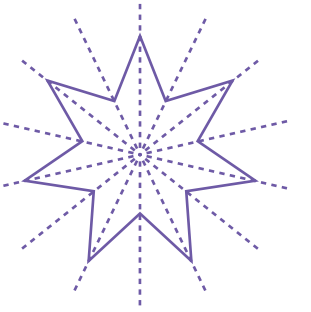
Worksheets
Symmetry

Symmetry of flat shapes

When a shape with **line symmetry** is folded along a line, one half of the shape will fit exactly on top of the other half. The fold line is called the **axis of symmetry**.

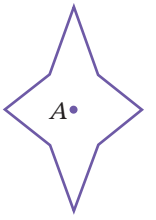


This ellipse has 2 axes of symmetry.

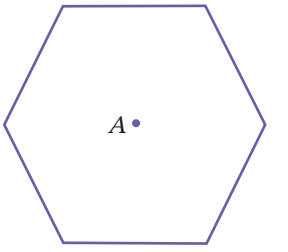


This star shape has 7 axes of symmetry.

When a shape with **rotational symmetry** is rotated about the point, called the **centre of symmetry**, it fits on itself at least once before one complete revolution.



This star shape fits on itself twice in one revolution, so it has rotational symmetry of order 2.



The regular hexagon fits on itself 6 times in one revolution, so it has rotational symmetry of order 6.

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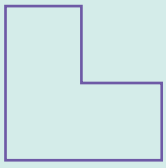
Example 11

For each shape:

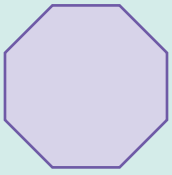
i count how many axes of symmetry it has

ii state whether it has rotational symmetry and the order if it does.

a



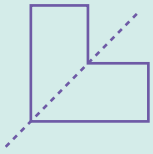
b



Solution

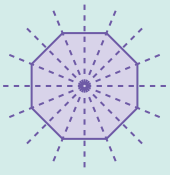
a **i** 1 axis of symmetry

ii no rotational symmetry



b **i** 8 axes of symmetry

ii rotational symmetry of order 8



EXERCISE 6.05 ANSWERS ON P. 626

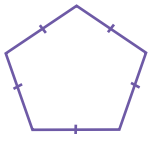
Line and rotational symmetry U F R C

1 Copy each shape and mark its axes of symmetry. R C

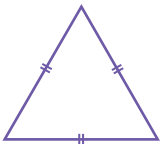
a



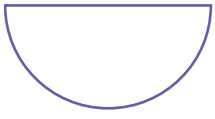
b



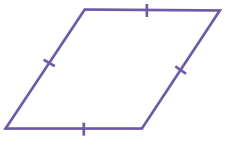
c



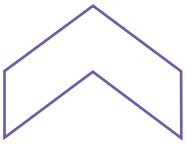
d



e



f

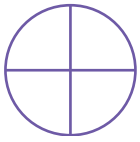


2 State whether each shape has rotational symmetry and the order if it does. R C

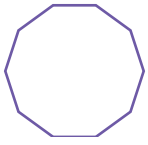
a



b



c



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PATH

3 Which of these capital letters have: R C

B C H M O P S W X Y Z

a line symmetry? **b** rotational symmetry?

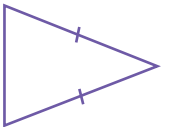
EXAMPLE 11

4 For each shape: R C

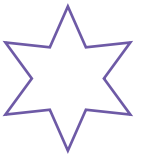
i find how many axes of symmetry it has

ii state whether it has rotational symmetry and the order if it does.

a



b



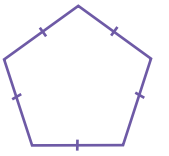
c



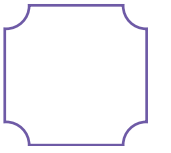
d



e



f



5 For each shape, find: R C

i the number of axes of symmetry

ii the order of rotational symmetry

a square **b** rhombus **c** regular decagon (10 sides)

d circle **e** kite

6 **a** Find the number of axes of symmetry and the order of rotational symmetry for:

i a scalene triangle **ii** an isosceles triangle **iii** an equilateral triangle

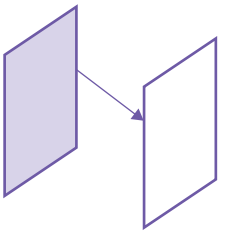
b What do you notice about your answers to part **a**? R C

6.06 Transformations on the number plane

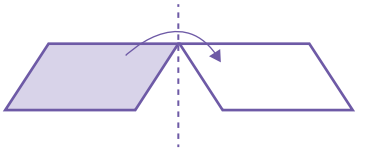
PATH

Worksheet
Transformations

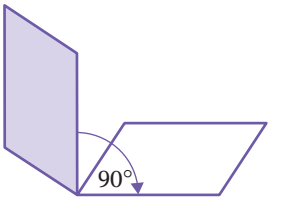
There are 3 types of **transformations**.



translation ('slide')



reflection ('flip')



rotation ('spin')

The shapes have changed position (transformed) but still have the same shape and size.

The original shape is called the **original**.

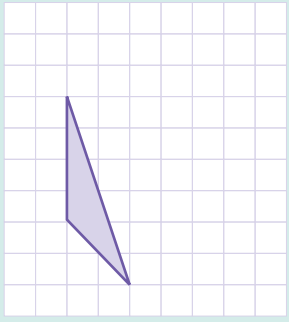
The transformed shape is called the **image**.

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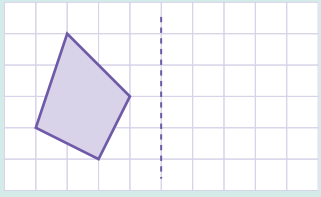
Example 12

a Translate this triangle 4 units right and 2 units up.



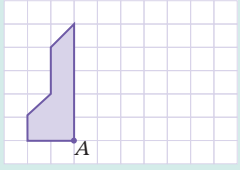
b Reflect the quadrilateral across the dotted line.

The dotted line is called the **line of reflection**.



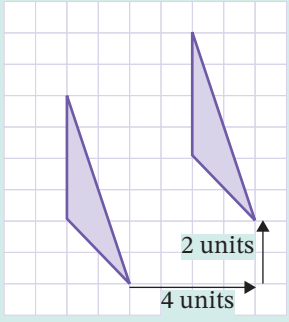
c Rotate the shape 90º in a clockwise direction about the point A.

A is called the **centre of rotation**.



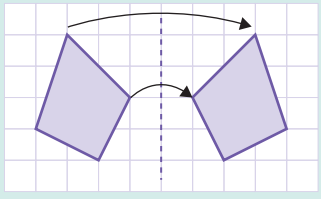
Solution

a



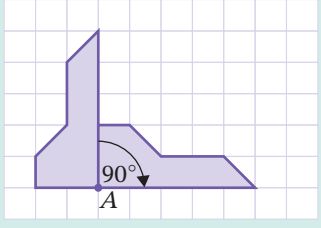
Every point on the original is shifted 4 units right and 2 units up.

b



The original and its mirror image are the same distance from the line of reflection.

c

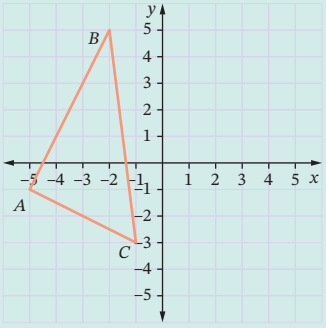


Every point on the original and image are the same distance from the centre of rotation.

PATH

When a point or vertex P of an original shape is transformed, the corresponding point or vertex on the image is labelled P', pronounced 'P-dash' or 'P-prime'.

Example 13



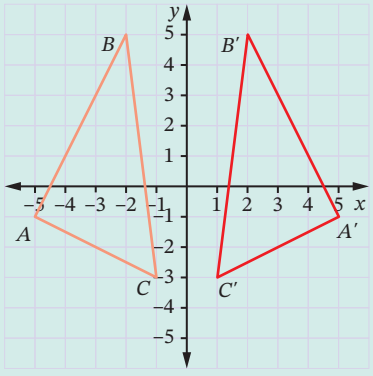
a Reflect the $\triangle ABC$ across the y-axis to obtain the image $\triangle A'B'C'$.

b Compare the coordinates of each vertex of $\triangle ABC$ to its matching vertex in $\triangle A'B'C'$.

c Now reflect $\triangle ABC$ across the x-axis to obtain the image $\triangle A'B'C'$. Compare the coordinatesof each vertex of the original to its matching vertex in its image.

Solution

a



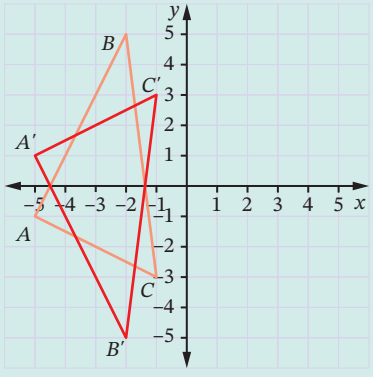
b $A\,(-5, -1) \rightarrow A'\,(5, -1)$

$B\,(-2, 5) \rightarrow B'\,(2, 5)$

$C\,(-1, -3) \rightarrow C'\,(1, -3)$

When reflected in the y-axis, the x-coordinate of each vertex changes sign while the y-coordinate stays the same.

c



$A\,(-5, -1) \rightarrow A'\,(-5, 1)$

$B\,(-2, 5) \rightarrow B'\,(-2, -5)$

$C\,(-1, -3) \rightarrow C'\,(-1, 3)$

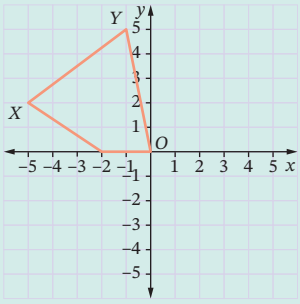
When reflected in the x-axis, the y-coordinate of each vertex changes sign while the x-coordinate stays the same.

Example 14

a Rotate $OWXY$ clockwise about the origin O through each angle to obtain $OW'X'Y'$.

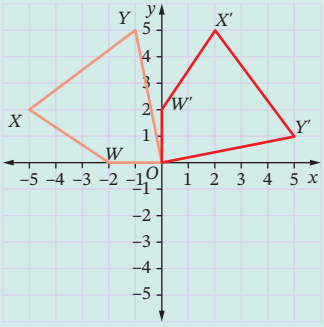
i 90° **ii** 180° **iii** 270°

b Compare the coordinates of each vertex of $OWXY$ to each vertex of the image $OW'X'Y'$ for each rotation in part **a**.



Solution

a, b i



$O\,(0, 0) \rightarrow O\,(0, 0)$

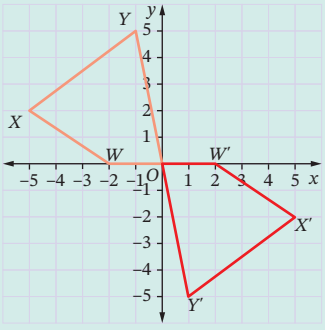
O is the centre of rotation

$W\,(-2, 0) \rightarrow W'(0, 2)$

$X\,(-5, 2) \rightarrow X'(2, 5)$

$Y\,(-1, 5) \rightarrow Y'(5, 1)$

ii



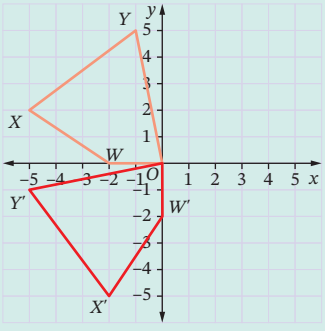
$O\,(0, 0) \rightarrow O\,(0, 0)$

$W\,(-2, 0) \rightarrow W'(2, 0)$

$X\,(-5, 2) \rightarrow X'(5, -2)$

$Y\,(-1, 5) \rightarrow Y'(1, -5)$

iii



$O\,(0, 0) \rightarrow O\,(0, 0)$

$W\,(-2, 0) \rightarrow W'(0, -2)$

$X\,(-5, 2) \rightarrow X'(-2, -5)$

$Y\,(-1, 5) \rightarrow Y'(-5, -1)$

PATH

Example 15

Find the new position of the point $M\,(4, 1)$ if it is:

a reflected in the x-axis.

b rotated anticlockwise through 90° about the origin.

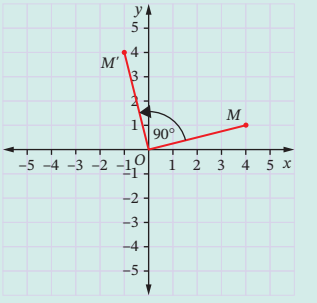
c translated 7 units left and 4 units down.

Solution

a M is reflected in the x-axis so the x-coordinate stays the same
$\therefore$ the new position of M is $(4, -1)$

b M is rotated anticlockwise about the origin
$\therefore$ the new position of M is $(-1, 4)$

c The new position of M is $(4 - 7, 1 - 4) = (-3, -3)$



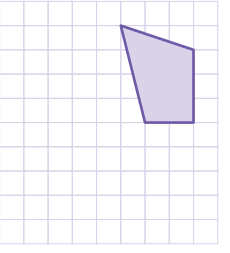
EXERCISE 6.06 ANSWERS ON P. 627

Transformations on the number plane UFRC

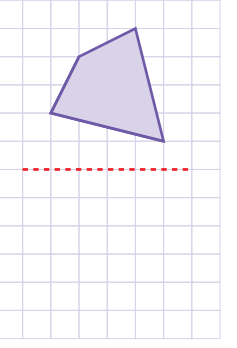
EXAMPLE 12

1 Copy each shape onto grid paper and draw the image after the stated transformation.

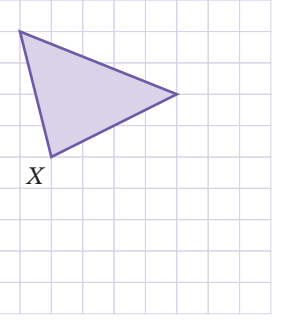
a Translate 3 units left and 4 units down.



b Reflect across the dotted line.

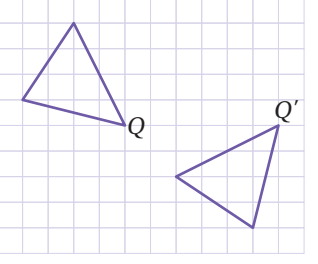


c Rotate 90° clockwise about X.

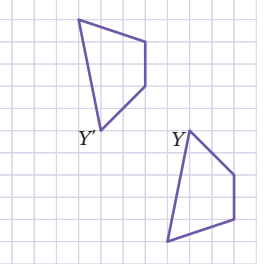


2 State the combination of 2 transformations used on the original to obtain the image.

a



b



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PATH

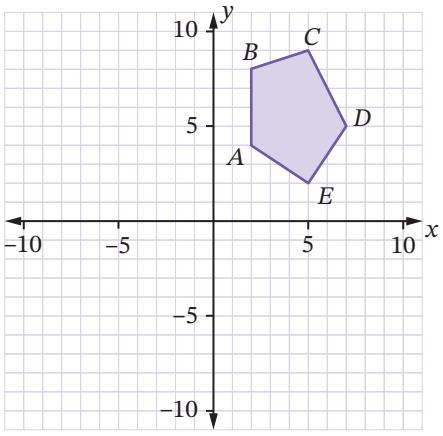
6.06

3 **a** Copy and reflect the shape *ABCDE* across the *y*-axis to obtain the image *A′B′CD′E′*.

b Compare the coordinates of each vertex of *ABCDE* to its matching vertex in *A′B′C′D′E′*.

c Now copy and reflect *ABCDE* across the *x*-axis to obtain the image *A′B′C′D′E′*.

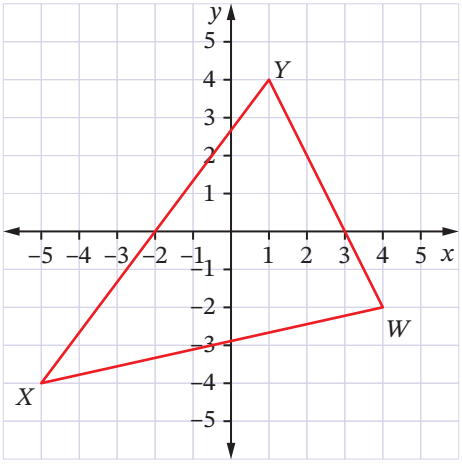
Compare the coordinates of each vertex of the original to its matching vertex in its image. R C



EXAMPLE 13

4 **a** Copy and reflect the △*WXY* across the *x*-axis to obtain the image △*W′X′Y′*.

b Write the coordinates of each vertex of △*W′X′Y′* and compare them to the coordinates of the vertices of △*WXY*. R C

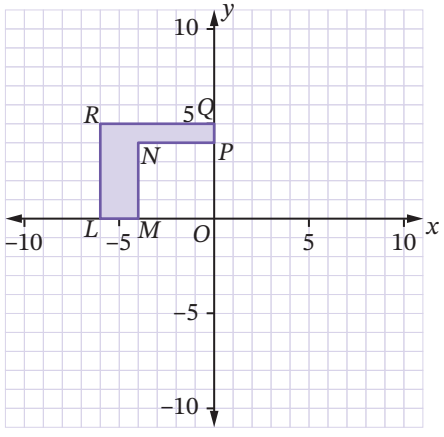


5 **a** Copy and rotate the shape *LMNPQR* clockwise about *O* through 90° to create the image *L′M′N′P′Q′*.

b Write down the coordinates of *P′* and *R′* and compare them to the coordinates of *P* and *R*.

c Write down the new positions of *M*, *N* and *Q* if *LMNPQ* is rotated about *O* through. R C

i 180° **ii** 270°



EXAMPLE 14

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PATH

6 **a** Copy and rotate $\triangle MPT$ clockwise about M through the following angles to obtain $\triangle M'P'T'$:

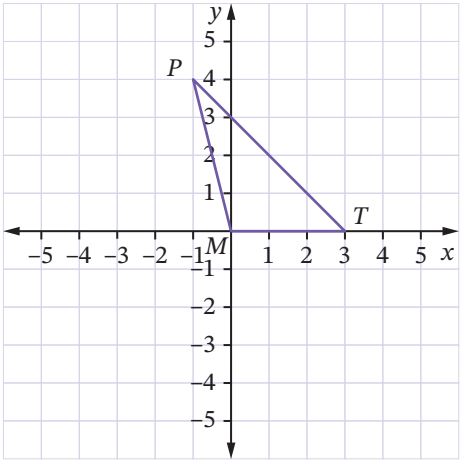
i 90° **ii** 180° **iii** 270°

b Compare the coordinates of the vertices of $\triangle MPT$ to those of $\triangle M'P'T'$ for each of the rotations. R C

c Copy and rotate $\triangle MPT$ anticlockwise about M through the following angles

i 90° **ii** 180° **iii** 270°

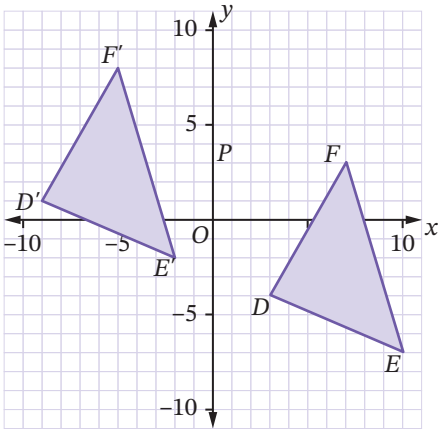
d Which of the images from part **c** are the same as the images from part **a**?



7 **a** Describe the transformation that has been performed on $\triangle DEF$.

b Compare the coordinates of each vertex of $\triangle DEF$ to its matching vertex in $\triangle D'E'F'$.

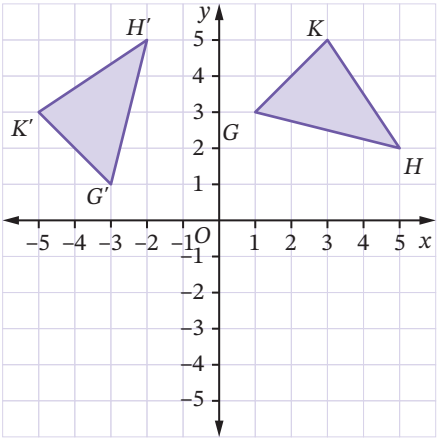
c Reflect $D'E'F'$ in the x-axis. Compare each vertex of the image of $\triangle D'E'F'$ to its matching vertex of $\triangle DEF$. R C



8 **a** Describe the transformation that has been performed on $\triangle GHK$.

b Compare the coordinates of each vertex of $\triangle GHK$ to its matching vertex in $\triangle G'H'K'$.

c Reflect $\triangle G'H'K'$ in the x-axis. Compare each vertex of the image of $\triangle G'H'K'$ to its matching vertex of $\triangle GHK$. R C



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PATH

EXAMPLE 15

6.06

9 Find the new position of the point $K(-5, 3)$ if it is: R C

a reflected in the x-axis. **b** reflected in the y-axis.

c rotated clockwise through 90° about the origin.

d translated 5 units left and 3 units up.

10 What are the coordinates of the image point P' if $P(-3, -7)$ is: R C

a reflected in the y-axis **b** reflected in the x-axis

c translated 5 units right and 4 units up

d rotated clockwise about the origin through

i 90° **ii** 180° **iii** 270°

11 Find the new position of the point $W(2, 3)$ if it is: R C

a translated 5 units right and 4 units down.

b rotated through 180° about the origin. **c** reflected in the y-axis.

d reflected in the x-axis and then translated 4 units left and 2 units down.

e rotated clockwise through 270° about the origin and then reflected in the x-axis.

12 $A(-4, 5)$ is reflected in the y-axis and then translated 5 units left and 2 units down. What are the new coordinates of the point A? Select the correct answer **A, B, C** or **D**.

A (−9, 3) **B** (−1, 3) **C** (−1, −3) **D** (9, −3)

13 What are the coordinates of the image point X′ if X (−4, −8) has been rotated about the origin through 180° and then reflected in the x-axis. Select **A, B, C** or **D**.

A (−4, 8) **B** (4, 8) **C** (8, −4) **D** (4, −8)

14 a The point $A(3, -4)$ is the image of a point after it has been translated 5 units right and 3 units down. Find the original point. R C

b The point G (−2, 5) is the image of a point after it been rotated 90° about the origin O. Find the original point.

c Find the original point if (6, −1) is the image of a point after it has been translated 2 units left and 4 units down and then reflected in the y-axis.

Technology

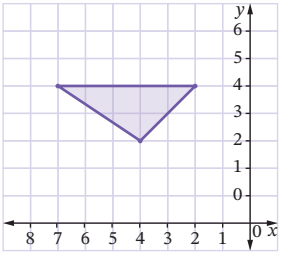
Transformations on the number plane

In this activity you will create reflections, rotations and translations of various shapes using **dynamic geometry software.** Create a number plane that shows all 4 quadrants and values of x and y from –10 to 10.

Activity 1: Reflection

1 Draw a **triangle** with vertices at (–7, 5), (–2, 5) and (–4, 3).

2 Draw a **reflection line on the *y*-axis,** through the points (0, 6) and (0, 0).



3 Reflect the triangle across the line.

4 Click a vertex and drag it. What happens?

5 Draw a new **reflection line** through the points (2, 7) and (–5, –2).

6 Reflect the triangle across this line.

Activity 2: Rotation

1 Draw a **polygon** with vertices at (0, 3), (–2, 3), (–3, 2), (–2, 0) and (0, 3).

2 Select the polygon and rotate it 45° clockwise about the point (0, 3).

3 Repeat step **2**, then rotate the polygon 180° clockwise about (0, 3). What happens?

4 Try other points as the centre point and enter different angles. What happens?

Activity 3: Translation

1 Draw a **polygon** with vertices at (–3, 4), (–3, –1), (1, –1), (1, 1), (–2, 1), (–2, 4) and (–3, 4).

2 Translate the polygon so that the image of (1, 1) is (4, 3).

3 Now translate the shape in 2 other ways.

Activity 4

1 Construct a **Polygon**.

2 Reflect the polygon about a point.

3 Reflect the polygon about a line.

4 Rotate the polygon by an angle.

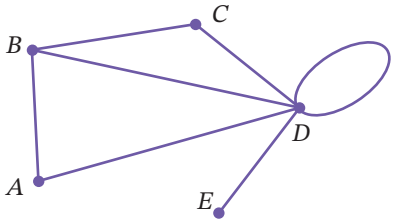
5 Translate the polygon by a vector.

Networks

6.07

A **network** is a collection of objects connected to each other in some way or system. Examples of networks include bus, rail and air routes, computer systems and family or friend relationships.

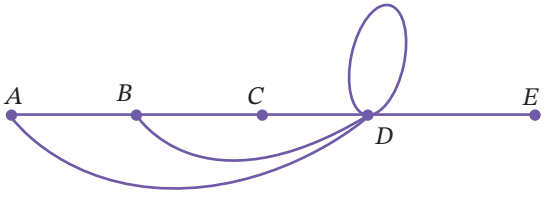
A **graph** is a diagram of a network where the objects are represented by points called **vertices** (or **nodes**), which are joined by lines called **edges**. The study of networks is called **graph theory**.



This graph or network diagram has 5 vertices, *A, B, C, D, E,* and 7 edges, *BC, BD, BA, AD, ED, CD, DD.*

A **loop** is an edge that connects a vertex to itself, such as *DD* in the diagram.

A network can be drawn in different ways as long as it shows the same connections. This diagram represents the same network as above.



PATH

Videos
Networks
The seven bridges of Konigsberg
Networks: Labyrinths and mazes

6.07

Degree of a vertex

The **degree** (or **order**) of a vertex is the number of edges connected to it, with loops counted as 2 edges. An **even vertex** has an even number of edges connected to it. An **odd vertex** has an odd number of edges.

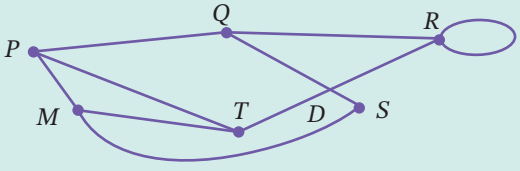
The **sum of the degrees** of all vertices in a network is equal to **twice the number of edges**, because each edge belongs to 2 vertices. This means that the sum of the degrees in a network is always even.

Example 16

a Find the degree of each vertex in this network.

b How many vertices are odd?

c Show that the sum of the degrees is equal to twice the number of edges.



Solution

a

Vertex	*M*	*P*	*Q*	*R*	*S*	*T*
Degree	3	3	3	4	2	3

b There are 4 odd vertices.

c Number of edges = 9.

Sum of degrees of vertices = 3 + 3 + 3 + 4 + 2 + 3 = 18,

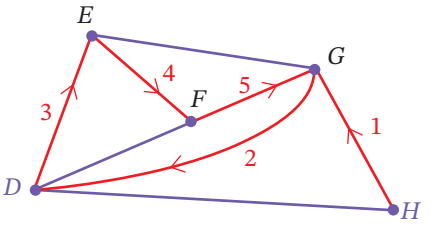
which is twice the number of edges (2 × 9).

PATH

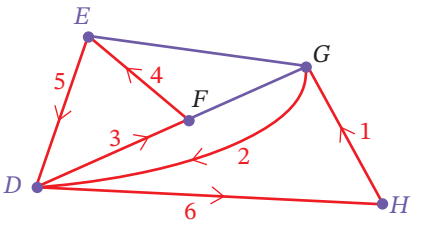
Walks, trails, paths and cycles

A **walk** is *any* route along the edges of a network, and may include repeated vertices or edges. Any walk that starts and ends at the same vertex is called a is **closed walk**.

A **trail** is a walk with no repeated edges, such as shown in the network below. However, a vertex can be repeated, such as *G* in *HGDEFG*. A **circuit** is a **closed trail**, such as *HGDFEDH* shown below (starts and ends at *H*).



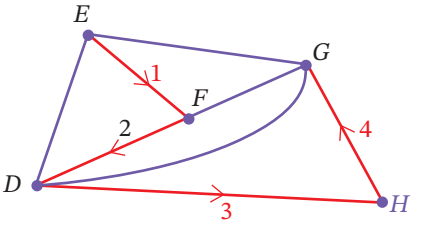
trail



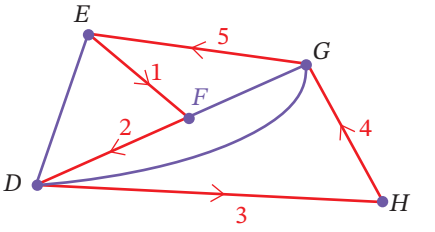
circuit

Think of a bushwalker walking on a trail or circuit and not repeating an edge.

A **path** is a walk with no repeated vertices (or edges), such as *EFDHG* shown in the network below. A **cycle** is a **closed path**, such as *EFDHGE* shown below (starts and ends at *E*, no repeated vertices).



path



cycle

Think of a cyclist riding along a bicycle path or cycle and not revisiting any points.

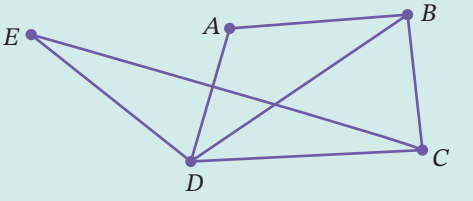
Example 17

a State whether each walk is a trail or path.

i *BCED* **ii** *ADECBDC*

b Write down a circuit that uses the edge *DC*.

c Write down 2 cycles that use the edge *DC*.



Solution

a **i** *BCED* is a path since no vertices (and edges) are repeated.

ii *ADECBDC* is a trail as no edges are repeated but vertex *D* is repeated.

b *ABDCEDA* is a circuit because it starts and ends at *A* with no repeated edges.

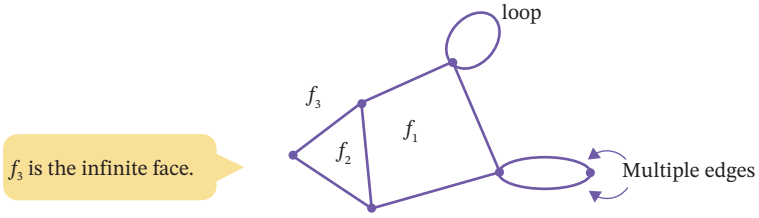
c *ADCBA* and *CEDC* are cycles because they are closed and have no repeated vertices.

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Faces of a network

A **face** of a network is a region bounded by edges. Regions formed by loops or multiple edges between the same vertices are *not* faces, but the external (outside) region of a network is a face. The external face is called the **infinite face**.

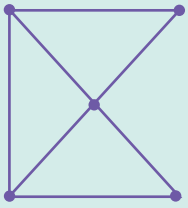
This network has 3 faces, f_1, f_2, f_3.



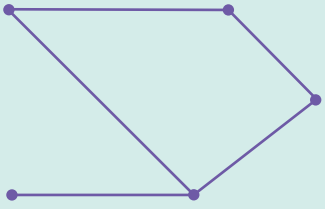
Example 18

Count the number of faces in each network.

a

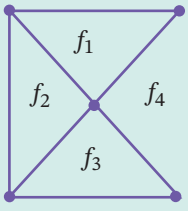


b



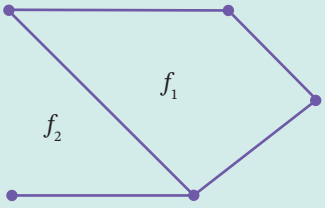
Solution

a



4 faces

b



2 faces

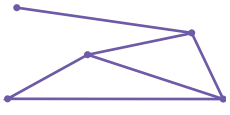
EXERCISE 6.07 ANSWERS ON P. 629

Networks UFRC

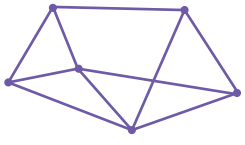
1 For each network, count the number of:

i vertices **ii** edges

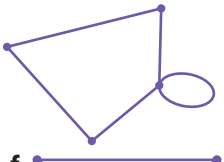
a



b



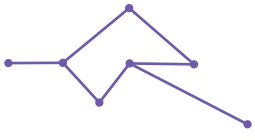
c



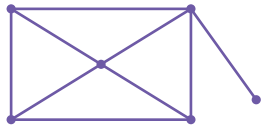
d



e



f



EXAMPLE 16

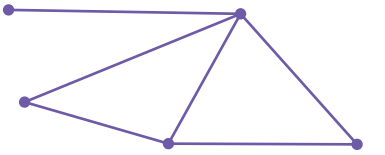
Foundation Standard Complex

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PATH

2 How many even vertices has this network?

Select the correct answer **A**, **B**, **C** or **D**.

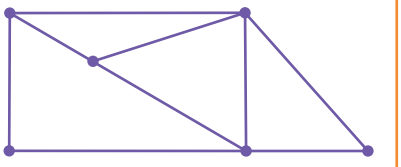


A 1 **B** 2

C 3 **D** 4

3 How many odd vertices has this network?

Select **A**, **B**, **C** or **D**.



A 2 **B** 3

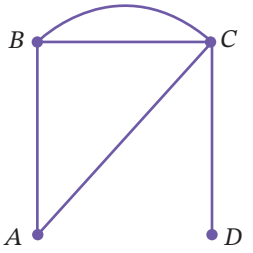
C 4 **D** 5

4 For each network: R C

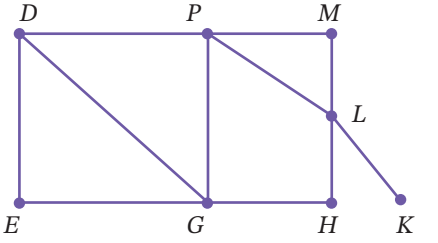
i find the degree of each vertex

ii show that the sum of the degrees is equal to twice the number of edges.

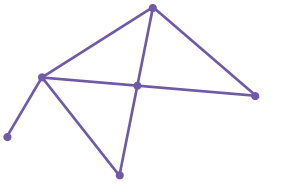
a



b



5 What is the sum of the degrees of all vertices in this graph? Select **A**, **B**, **C** or **D**.

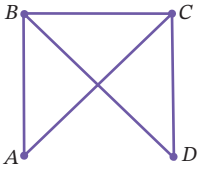


A 8 **B** 12

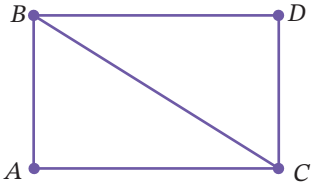
C 16 **D** 20

6 Which of these graphs does ***not*** show the same information as the given network diagram?

Select **A**, **B**, **C** or **D**. R C



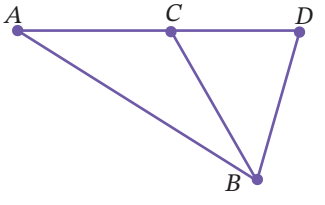
A



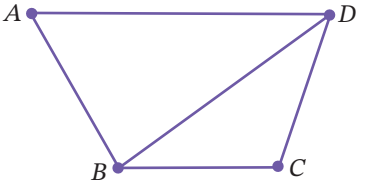
B



C



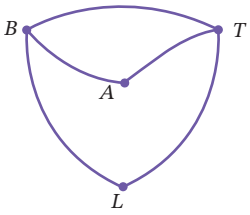
D



Foundation Standard Complex

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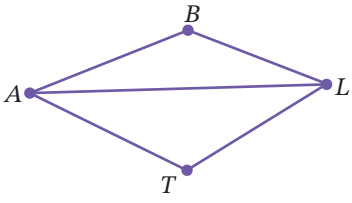
7 This network shows the friendship connections between Lara (L), Tasnuba (T), Aditi (A) and Belinda (B). An edge represents a friendship between 2 people.



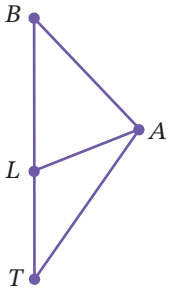
Which graph shows the same information?

Select **A**, **B**, **C** or **D**. R C

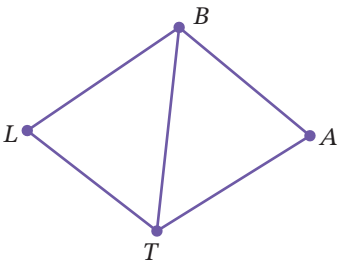
A



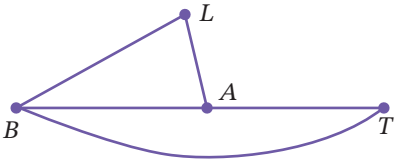
B



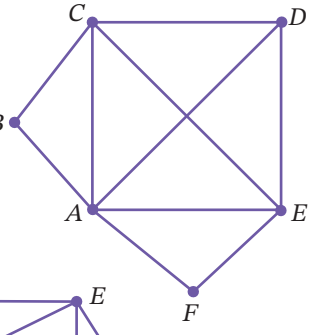
C



D



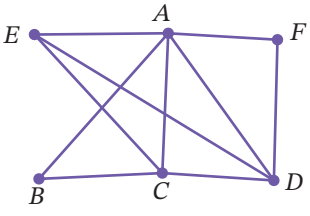
8 The graph shows the communication network between division managers in a business.



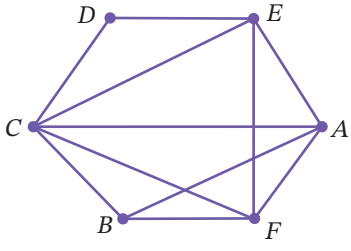
Which graph shows the same information?

Select **A**, **B**, **C** or **D**. R C

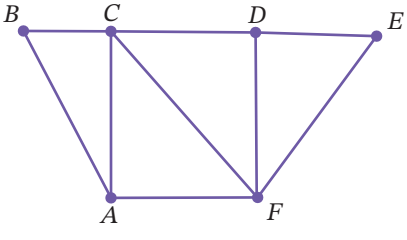
A



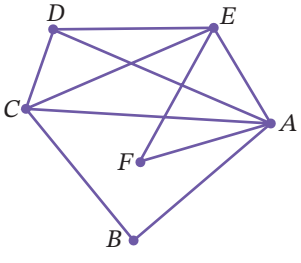
B



C



D



Foundation | Standard | Complex

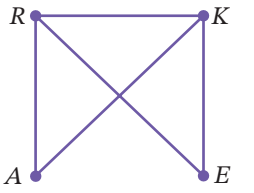
9780170479585

PATH

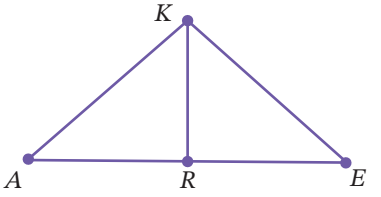
9 This graph shows social media connections between Ashley (A), Emily (E), Kristy (K) and Robyn (R).

Which graph shows the same information?

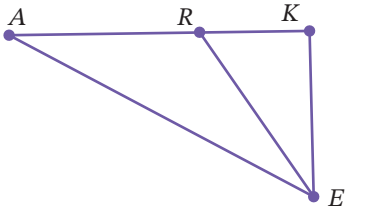
Select **A**, **B**, **C** or **D**. R C



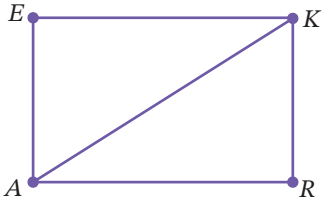
A



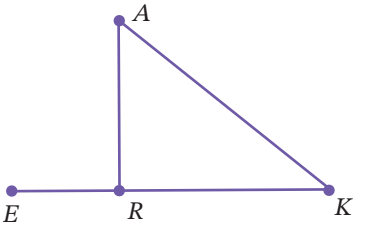
B



C

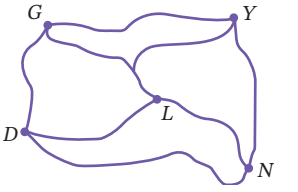


D

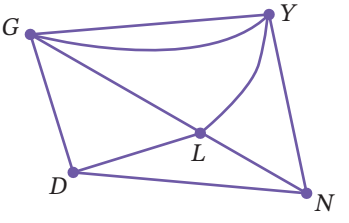


10 Which network diagram represents this map showing the roads that connect 5 towns?

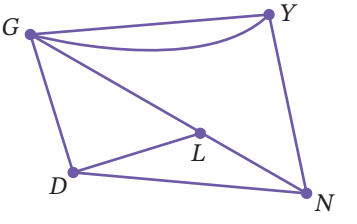
Select **A**, **B**, **C** or **D**. R C



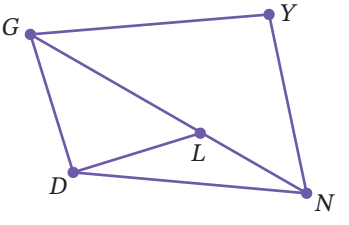
A



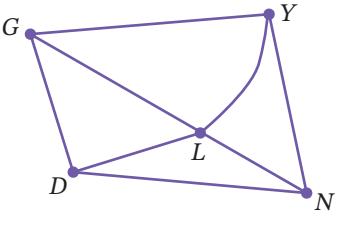
B



C

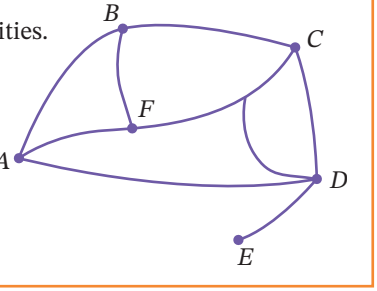


D



11 The diagram shows the airline connections between 6 cities.

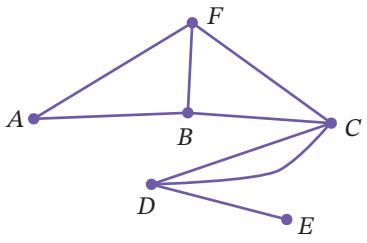
Which of the following represents the same information? Select **A**, **B**, **C** or **D**. R C



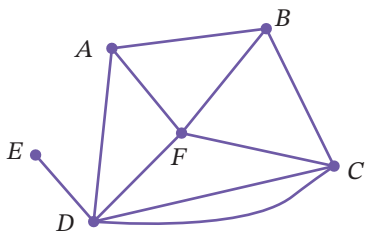
Foundation Standard Complex

9780170479585

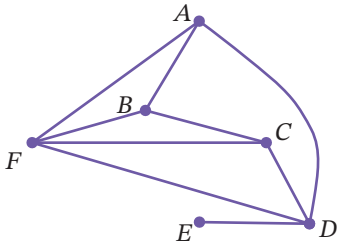
A



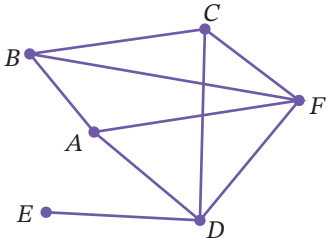
B



C



D



12 Arya, Cassie, Naved, Lachlan, Riya and Shayaan attend the same high school.

- Naved is friends with Arya, Shayaan and Riya
- Arya is friends with Naved and Riya
- Shayaan is friends with Naved, Riya, Lachlan and Cassie
- Lachlan is friends with Riya, Shayaan and Cassie
- Cassie is friends with Shayaan and Lachlan
- Riya is friends with Arya, Shayaan and Lachlan R C

a Draw a graph to represent the above network of friends.

b Who is not a friend with Riya?

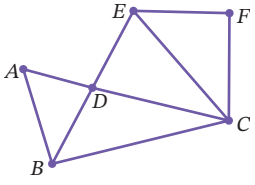
13 **a** State whether each walk is a trail or path. R C

i *ABDCED* **ii** *CFEDBC*

iii *FEDBA* **iv** *DECBA*

v *BDCFEDAB* **vi** *BDCB*

EXAMPLE 17



b Which of the walks in part **a** is:

i a circuit (a closed trail)? **ii** a cycle (a closed path)?

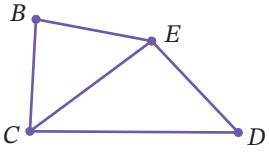
c Write down a circuit that contains the edge *DC*.

d Write down a cycle that contains the edges *AB* and *FE*.

14 Which of the following is a trail? Select **A**, **B**, **C** or **D**. R C

A *CEDCB* **B** *DECBEC*

C *ECBDE* **D** *CBECDEB*

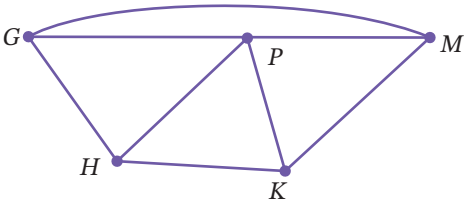


Foundation Standard Complex

PATH

15 Which of the following is a circuit?

Select **A**, **B**, **C** or **D**. R C



A *HPKMPH*

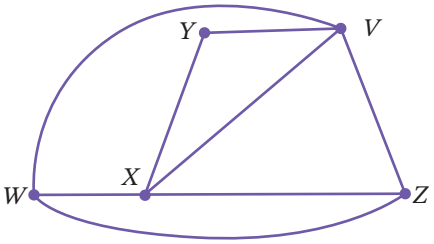
B *GHPKMPG*

C *PKMGHKM*

D *KHGPHKMG*

16 Which of the following is a cycle?

Select **A**, **B**, **C** or **D**. R C



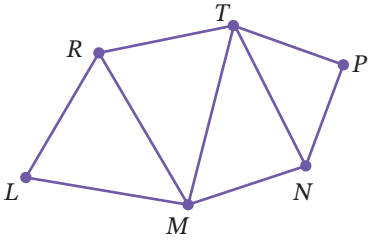
A *ZVXYVZW*

B *ZXYVXZW*

C *YXZWVY*

D *ZXVWY*

17 For the network diagram shown, write down: R C



a a trail that includes the edge *MN*

b a walk that includes the edges *PT* and *RM*

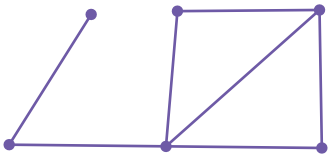
c a circuit that begins at *L* and includes edge *NT*

d a walk that begins at *T* and includes the edge *RM*

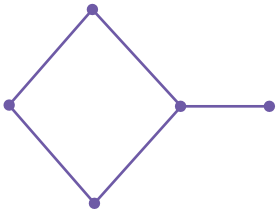
EXAMPLE 18

18 Count the number of faces in each network.

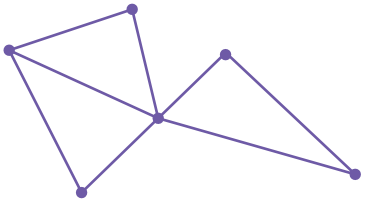
a



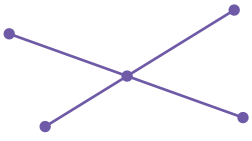
b



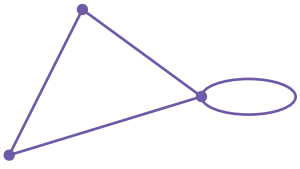
c



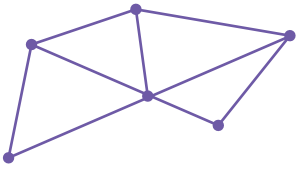
d



e



f



Foundation Standard Complex

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PATH

6.07

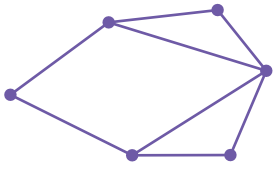
Investigation

Euler's rule

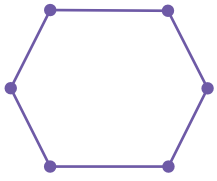
Copy and complete this table using each network below.

	Number of vertices, v	Number of faces, f	$v + f$	Number of edges, e
1				
2				
3				
4				
5				
6				
7				
8				
9				

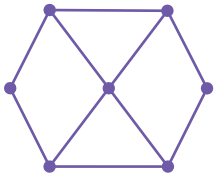
1



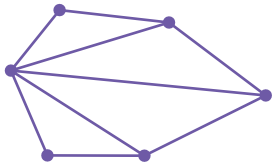
2



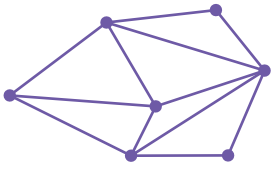
3



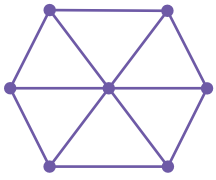
4



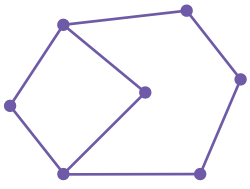
5



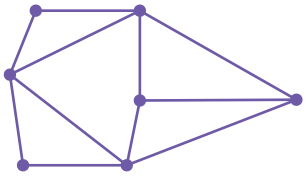
6



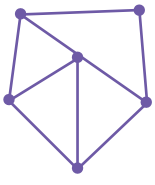
7



8



9



In 1758, the Swiss mathematician Leonhard Euler (pronounced 'oiler') discovered a relationship between $v + f$ (number of vertices and faces) and e (number of edges) for these networks.

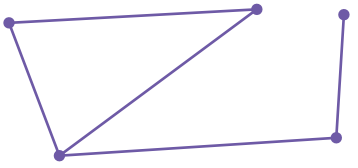
What was Euler's rule?

6.08 Planar graphs and Euler's formula

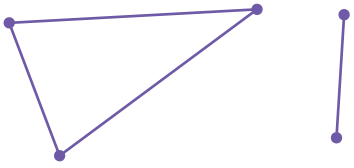
PATH

Video
Euler's formula for planar graphs

A network or graph is **connected** if there is a path between any 2 vertices on the network.

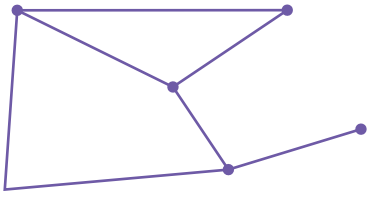


Connected

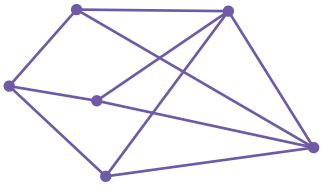


Not connected

A **planar graph** is a connected graph that can be drawn so that no edges cross over.
'Planar' means 2D, on a flat plane, and a planar graph can be drawn on a flat plane so that its edges don't overlap.

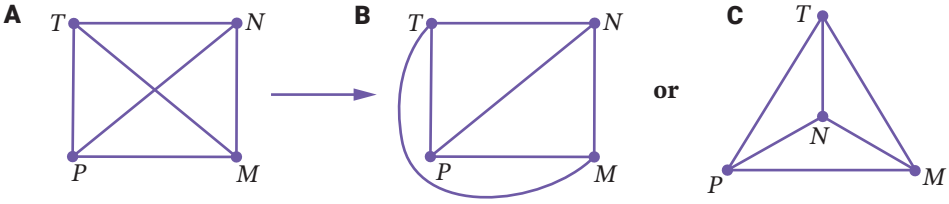


Planar graph



A non-planar graph

Some graphs may have edges crossing but they are planar graphs because they can be *redrawn* without any of the edges crossing. For example, Graph **A** below is planar because it can be redrawn as **B** or **C**.



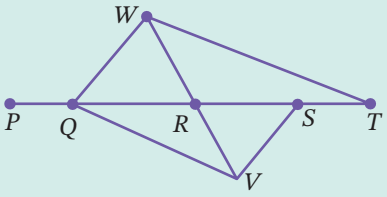
A **non-planar graph** has edges crossing and cannot be drawn so that the edges do not cross.

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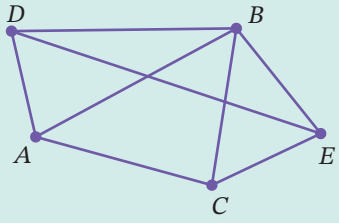
Example 19

State whether each network is a planar or non-planar graph.

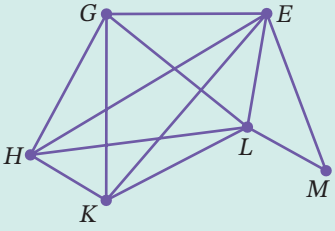
a



b



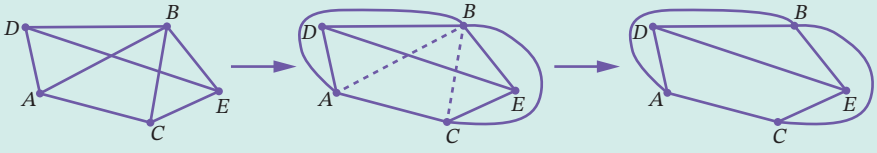
c



Solution

a A planar graph because no edges cross.

b The graph can be redrawn by moving AB and BC as shown.

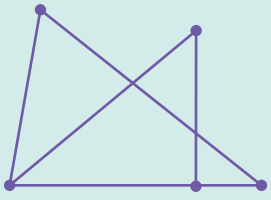


There are now no edges crossing so it's a planar graph.

c It is impossible to move the edges so that no edges cross over, so this is a non-planar graph.

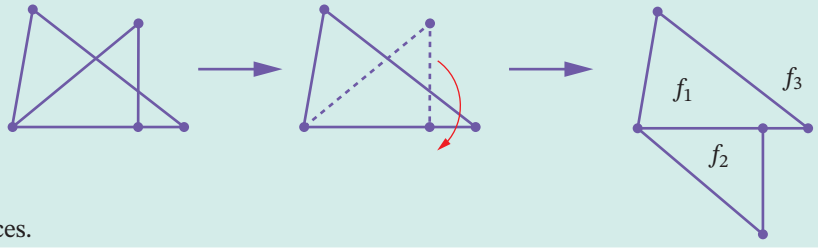
Example 20

Count the number of faces in this planar graph.



Solution

First redraw the graph so that no edges cross.



3 faces.

PATH

Euler's formula

The Swiss mathematician Leonhard Euler (pronounced 'oiler') discovered a relationship between the vertices, edges and faces of a planar graph.

Euler's formula for planar graphs

For a graph to be planar,

$$v - e + f = 2$$

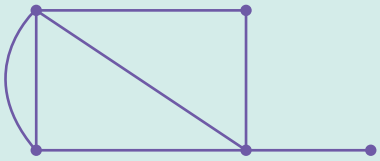
where v = number of vertices, e = number of edges, f = number of faces

Example 21

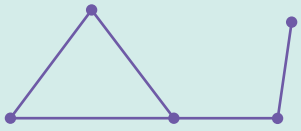
Verify Euler's formula for each planar graph.

'Verify' means to show that something is true or correct.

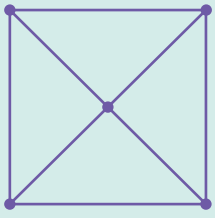
a



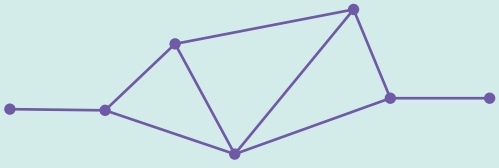
b



c



d



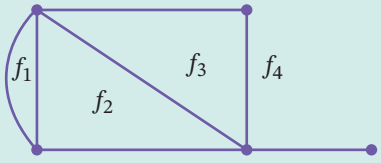
Solution

a $v = 5, e = 7$ and $f = 4$

$\therefore v - e + f = 5 - 7 + 4$

$= 2$

$\therefore$ Euler's formula is true.

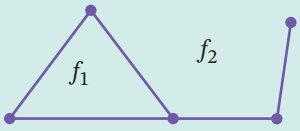


b $v = 5, e = 5$ and $f = 2$

$\therefore v - e + f = 5 - 5 + 2$

$= 2$

$\therefore$ Euler's formula is true.

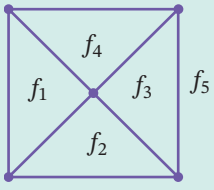


c $v = 5, e = 8$ and $f = 5$

$\therefore v - e + f = 5 - 8 + 5$

$= 2$

$\therefore$ Euler's formula is true.

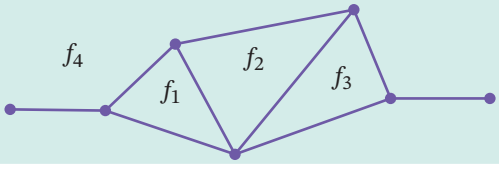


d $f = 4, v = 7$ and $e = 9$

$\therefore v - e + f = 7 - 9 + 4$

$= 2$

$\therefore$ Euler's formula is true.



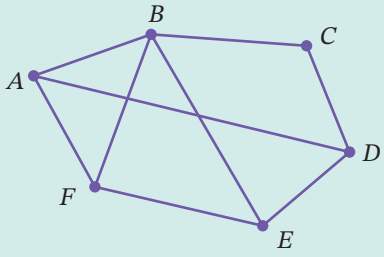
9780170479585

Example 22

Show that this graph is planar by:

a redrawing it so that no edges cross

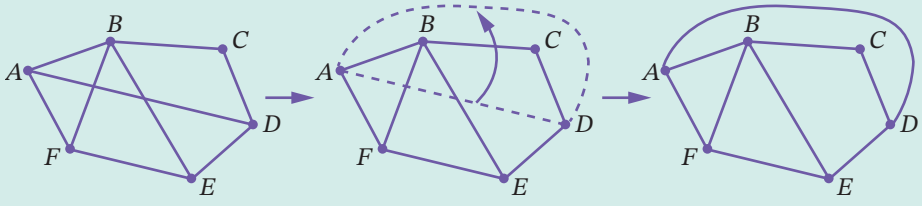
b using Euler's formula



Solution

a Edge AD crosses edges BF and BE, so move edge AD to the outside of vertices B and C.

Since the graph can redrawn without any edges crossing, it is a planar graph.

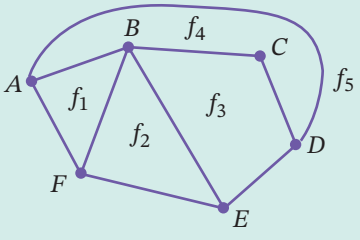


b $v = 6$, $e = 9$ and $f = 5$

$\therefore v - e + f = 6 - 9 + 5$

$= 2$

$\therefore$ The graph is planar.

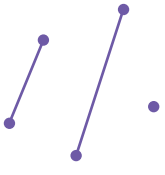


EXERCISE 6.08 ANSWERS ON P. 629

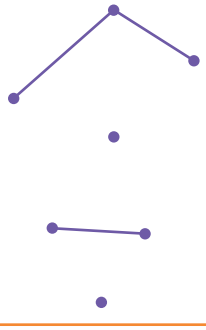
Planar graphs and Euler's formula U F R C

1 What is the smallest number of edges required to make each graph a connected network? R C

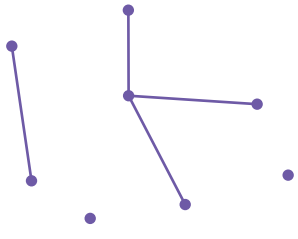
a



b



c



Foundation Standard Complex

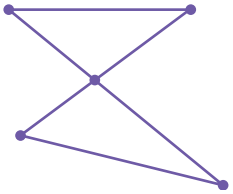
9780170479585

PATH

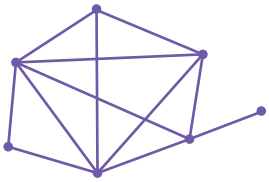
EXAMPLE 19

2 State whether each network is a planar or non-planar graph. Give a reason for your answer. C

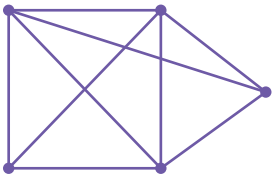
a



b

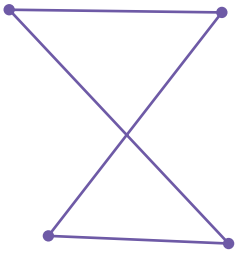


c

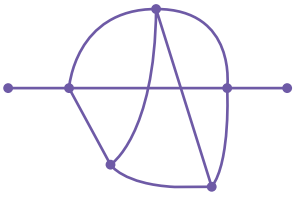


3 Redraw each graph to show that it is planar. C

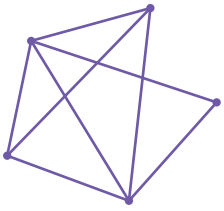
a



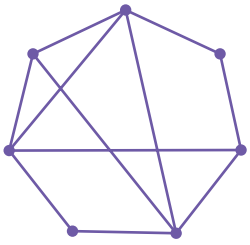
b



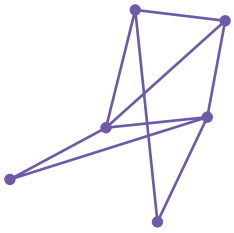
c



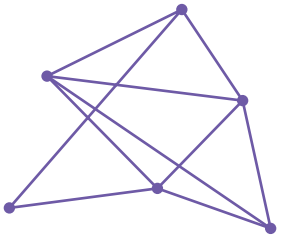
d



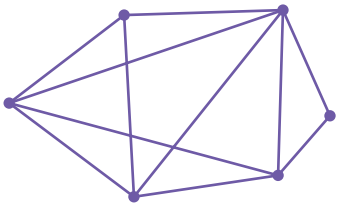
e



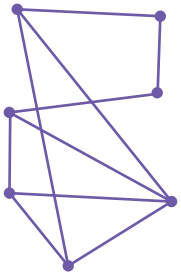
f



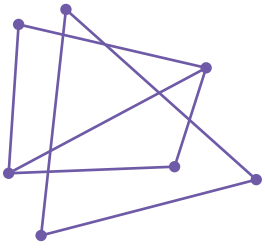
g



h



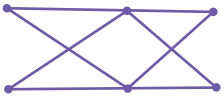
i



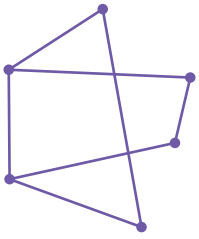
EXAMPLE 20

4 Redraw each planar graph to count its number of faces. R C

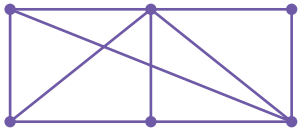
a



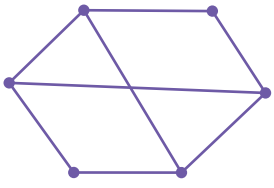
b



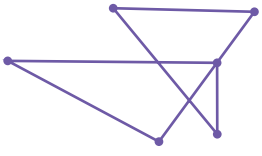
c



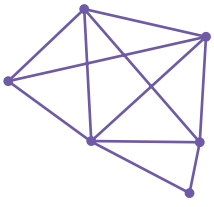
d



e



f



Foundation Standard Complex

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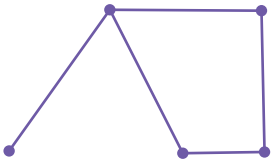
PATH

EXAMPLE 21

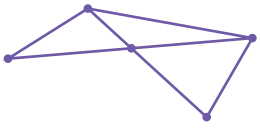
6.08

5 Verify Euler's formula for each planar graph. R C

a



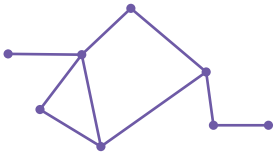
b



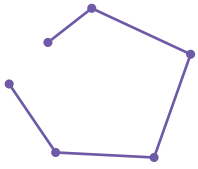
c



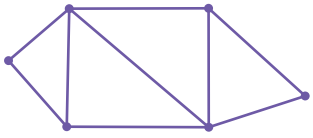
d



e

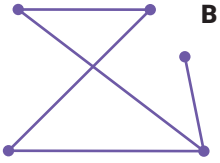


f

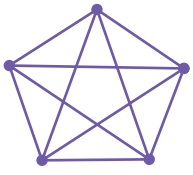


6 For which graph does Euler's formula NOT apply? R C

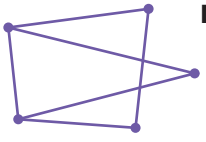
A



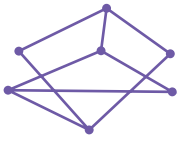
B



C



D

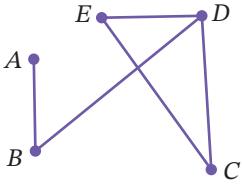


EXAMPLE 22

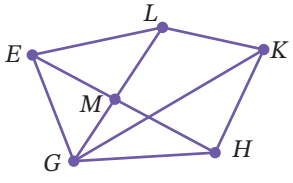
7 Show that each graph is planar by: R C

i redrawing it so that no edges cross **ii** using Euler's formula

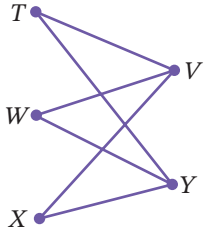
a



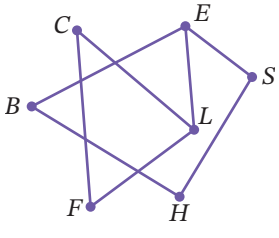
b



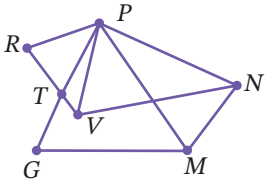
c



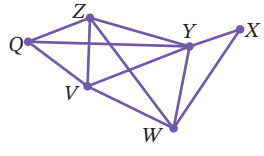
d



e



f



8 A planar graph has 14 edges and 8 faces. How many vertices has it?

Select the correct answer **A**, **B**, **C** or **D**.

A 5 **B** 6 **C** 7 **D** 8

9 A planar graph has 6 edges and 5 vertices. What is the number of faces?

Select **A**, **B**, **C** or **D**.

A 1 **B** 3 **C** 5 **D** 7

Foundation Standard Complex

PATH

10 A planar graph has 12 vertices and 7 faces. What is the number of edges?

Select **A**, **B**, **C** or **D**.

A 20 **B** 17 **C** 14 **D** 11

11 A planar graph has 9 faces. Which of the following could be its number of vertices and edges?

Select **A, B**, **C** or **D**.

A 6 vertices and 12 edges **B** 10 vertices and 7 edges

C 9 vertices and 13 edges **D** 11 vertices and 18 edges

Power plus ANSWERS ON P. 630

1 How many diagonals has:

a a quadrilateral? **b** an octagon? **c** a dodecagon?

2 Name all quadrilaterals whose diagonals:

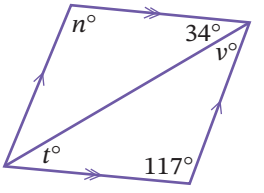
a bisect each other at right angles **b** bisect each other

c intersect at right angles **d** have equal length

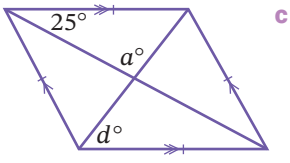
e bisect the angles of the quadrilateral **f** are equal and bisect each other

3 Find the value of each variable, giving reasons.

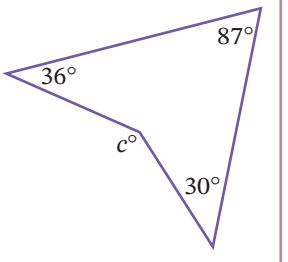
a



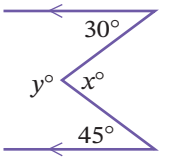
b



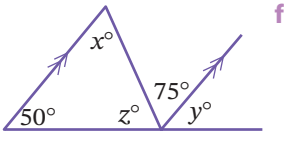
c



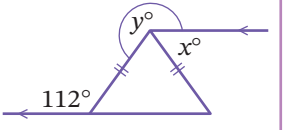
d



e

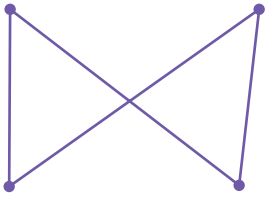


f

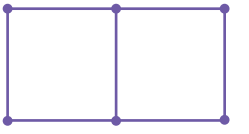


4 Draw each network without lifting your pen or pencil (drawing each edge only once). Check your work with that of other students.

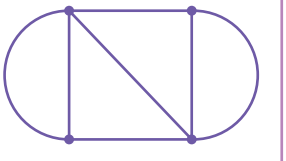
a



b



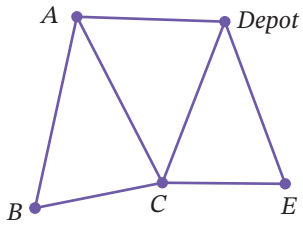
c



Foundation Standard Complex

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5 A roadworks crew is line marking along some roads as shown in the network diagram. The crew starts at the depot and wants to go along each road only once.



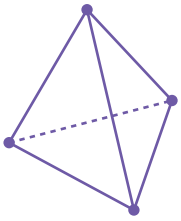
a Is it possible for the crew to go along each road only once?

b Is it possible for the crew to start and end at the depot?

6 **a** Redraw each solid as a (flat) network diagram.

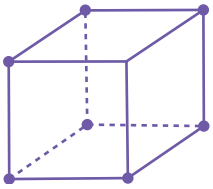
i tetrahedron

a



ii cube

b



b Are the networks of the solids planar graphs?

6.08

CHAPTER 6 REVIEW

Quiz
Language of maths 6

Language of maths

angle sum	convex	degree	edge
equilateral	Euler	exterior angle	face
isosceles	kite	line symmetry	network
parallelogram	planar graph	quadrilateral	reflection
regular polygon	rhombus	rotation	rotational symmetry
transformation	translation	trapezium	vertex (node)

1 What shape has 2 pairs of equal adjacent sides?

2 Name one property of a convex quadrilateral.

3 What type of angle is created when one side of a triangle is extended?

4 What is the degree of a vertex in a network?

5 What is a **regular polygon?** Are all regular polygons also **convex**?

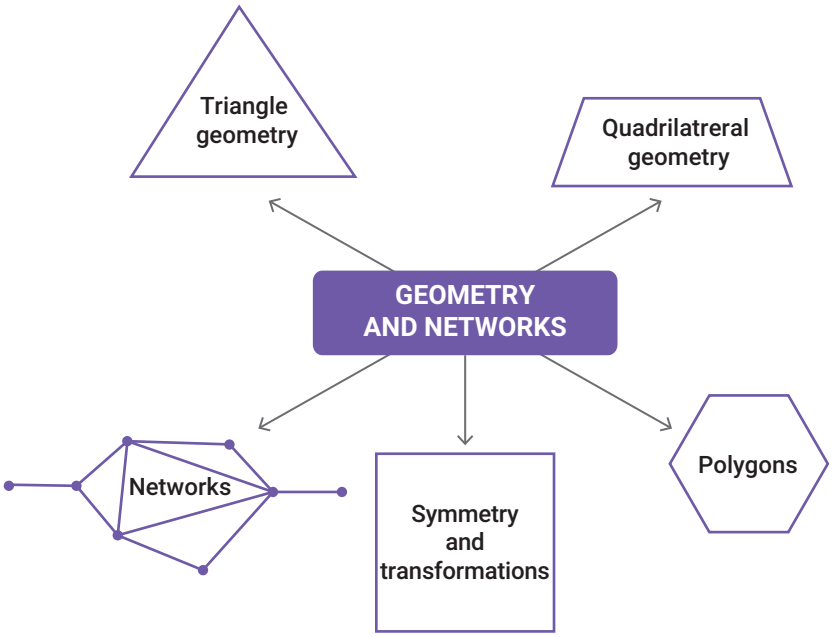
6 Write Euler's formula for planar graphs.

Topic summary

Worksheet
Mind map: Geometry and networks (Advanced)

- Write 3 ideas from this topic that were new to you.
- Name the types of triangles, choose one and list all of its properties.
- Name the 6 special quadrilaterals, choose one and list all of its properties.
- Summarise what you know about the features of a network.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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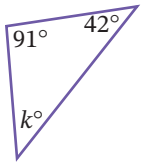
TEST YOURSELF 6

ANSWERS ON P. 630

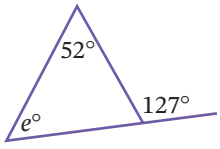
Quiz
Test yourself 6

1 Find the value of each variable, giving reasons. 6.01

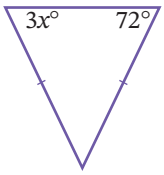
a



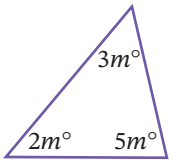
b



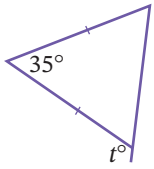
c



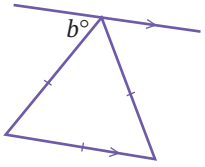
d



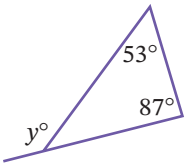
e



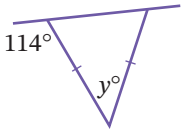
f



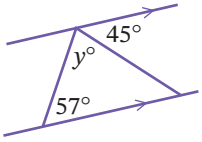
g



h



i

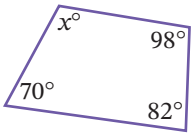


2 What are the sizes of the angles in: 6.01

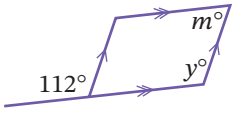
a an equilateral triangle? **b** a right-angled isosceles triangle?

3 Find the value of each variable, giving reasons. 6.02

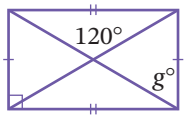
a



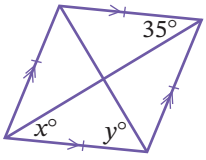
b



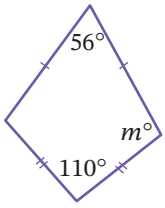
c



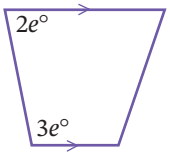
d



e



f



4 Name all quadrilaterals that have: 6.02

a both pairs of opposite sides parallel **b** 2 equal diagonals

c all sides equal **d** diagonals that bisect each other.

5 Draw: 6.03

a a regular pentagon **b** a non-regular pentagon **c** a concave pentagon

Foundation | Standard | Complex

PATH

6.03 6.04 6.05 6.06 6.06 6.07

6 **a** Show that the angle sum of a decagon is 1440°.

b Find the size of one angle in a regular nonagon.

7 **a** Find the size of each exterior angle in a regular hexagon.

b If each angle in a regular polygon is 150°, how many sides does it have?

8 For each shape, find:

i the number of axes of symmetry

ii the order of rotational symmetry

a rectangle **b** regular hexagon **c** rhombus

9 **a** Copy the shape $ABCD$ onto grid paper and draw its image after:

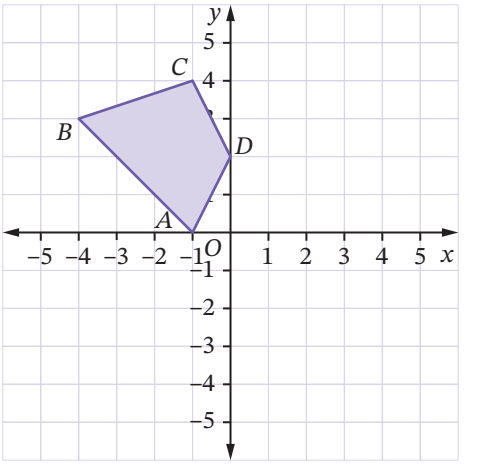
i reflecting it across the x-axis

ii reflecting it across the y-axis

iii rotating it clockwise about O through 90°.

iv translating it 4 units to the right and 3 units down.

b For each of the above transformations, compare the coordinates of each vertex of the original to its matching vertex in its image.



Worksheet
5 mm grid paper

10 Find the new position of the point $T(-3, -2)$ if it is:

a reflected in the x-axis.

b reflected in the y-axis.

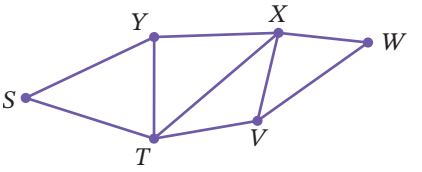
c rotated clockwise about O through 90°.

d translated 7 units right and 4 units up.

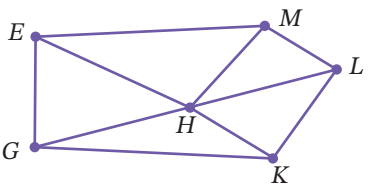
e rotated 180° about O and then reflected in the y-axis.

11 Find the degree of each vertex and show that the sum of the degrees is equal to twice the number of edges.

a



b



Foundation Standard Complex

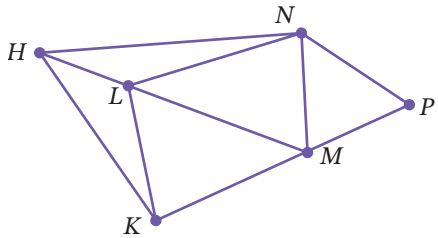
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12 **a** State whether each walk is a trail or a path.

i *HLNPML* **ii** *HKMNP*

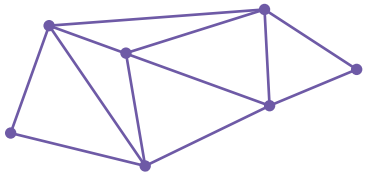
iii *HKMPNLH* **iv** *LNMLKM*

b Which of the walks in part **a** is a cycle?

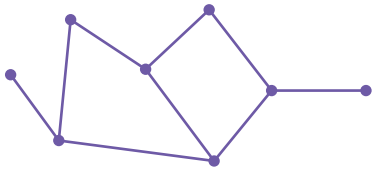


13 Verify Euler's formula for each planar graph.

a



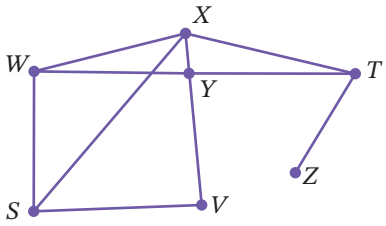
b



14 Show that each graph is planar by:

i redrawing it so that no edges cross **ii** using Euler's formula

a



b

